

ARM PrimeCell™[®]

UART (PL011)

Technical Reference Manual

The ARM logo, consisting of the letters "ARM" in a bold, sans-serif font.

ARM PrimeCell™

Technical Reference Manual

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Release Information

The following changes have been made to this document.

Change history		
Date	Issue	Change
July 12th 2000	A	First release
August 18th 2000	B	Change to signal names in Fig 2-1, changes to bits in Figs 4-1, 4-3.
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December 14th 2001	E	Text changes to pages 3-13, 3-14, and 3-17.

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Preface

This preface introduces the ARM PrimeCell UART (PL011) and its reference documentation. It contains the following sections:

- *About this document* on page vi
- *Further reading* on page viii
- *Feedback* on page ix.

About this document

This document is a technical reference manual for the ARM PrimeCell UART (PL011).

Intended audience

This document has been written for hardware and software engineers implementing System-on-Chip designs. It provides information to enable designers to integrate the peripheral into a target system as quickly as possible.

Using this manual

This document is organized into the following chapters:

Chapter 1 *Introduction*

Read this chapter for an introduction to the ARM PrimeCell UART (PL011).

Chapter 2 *Functional Overview*

Read this chapter for a description of the major functional blocks of the PrimeCell UART.

Chapter 3 *Programmer's Model*

Read this chapter for a description of the PrimeCell UART registers and programming details.

Chapter 4 *Programmer's Model for Test*

Read this chapter for a description of the logic in the PrimeCell UART for integration testing.

Appendix A *ARM PrimeCell UART (PL011) Signal Descriptions*

Read this appendix for details of the PrimeCell UART signals.

Typographical conventions

The following typographical conventions are used in this document:

bold	Highlights ARM processor signal names, and interface elements such as menu names. Also used for terms in descriptive lists, where appropriate.
<i>italic</i>	Highlights special terminology, cross-references and citations.
typewriter	Denotes text that can be entered at the keyboard, such as commands, file names, program names, and source code.

<u>typewriter</u>	Denotes a permitted abbreviation for a command or option. The underlined text can be entered instead of the full command or option name.
<i>typewriter italic</i>	Denotes arguments to commands or functions where the argument is to be replaced by a specific value.
typewriter bold	Denotes language keywords when used outside example code.

Timing diagram conventions

This manual contains one or more timing diagrams. The following key explains the components used in these diagrams. Any variations are clearly labeled when they occur. Therefore, no additional meaning should be attached unless specifically stated.

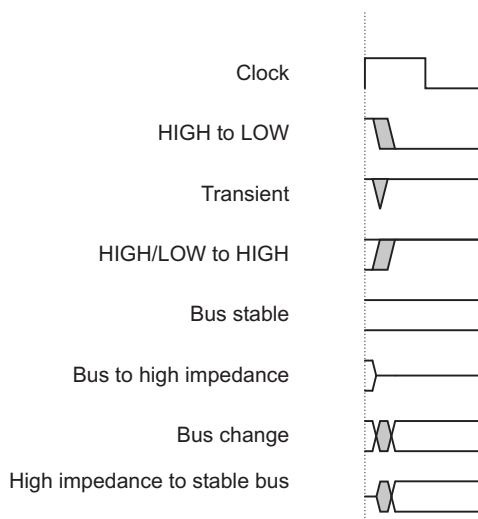


Figure P-1 Key to timing diagram conventions

Shaded bus and signal areas are undefined, so the bus or signal can assume any value within the shaded area at that time. The actual level is unimportant and does not affect normal operation.

Further reading

This section lists publications by ARM Limited, and by third parties.

ARM periodically provides updates and corrections to its documentation. See <http://www.arm.com> for current errata sheets and addenda.

See also the ARM Frequently Asked Questions list at:
<http://www.arm.com/DevSupp/Sales+Support/faq.html>

ARM publications

This document contains information that is specific to the ARM PrimeCell UART (PL011). Refer to the following documents for other relevant information:

- *AMBA Specification (Rev 2.0)* (ARM IHI 0011)
- *ARM PrimeCell UART (PL011) Design Manual* (PL011 DDES 0000)
- *ARM PrimeCell UART (PL011) Integration Manual* (PL011 INTM 0000).

Other publications

This section lists relevant documents published by third parties.

- *Infrared Data Association (IrDA) Serial Infrared Physical Layer Link Specification v1.1* (17 Oct 1995)
- *Hewlett-Packard IrDA data link design guide* (5964-0245E. Aug 1995).

Feedback

ARM Limited welcomes feedback both on the ARM PrimeCell UART (PL011), and on the documentation.

Feedback on the ARM PrimeCell UART (PL011)

If you have any comments or suggestions about this product, please contact your supplier giving:

- the product name
- a concise explanation of your comments.

Feedback on this document

If you have any comments on about this document, please send email to errata@arm.com giving:

- the document title
- the document number
- the page number(s) to which your comments refer
- a concise explanation of your comments.

General suggestions for additions and improvements are also welcome.

Chapter 1

Introduction

This chapter introduces the ARM PrimeCell UART (PL011). It contains the following section:

- *About the ARM PrimeCell UART (PL011)* on page 1-2.

1.1 About the ARM PrimeCell UART (PL011)

The PrimeCell UART is an *Advanced Microcontroller Bus Architecture* (AMBA) compliant *System-on-Chip* (SoC) peripheral that is developed, tested, and licensed by ARM.

The PrimeCell UART is an AMBA slave module, and connects to the *Advanced Peripheral Bus* (APB). The PrimeCell UART includes an *Infrared Data Association* (IrDA) *Serial InfraRed* (SIR) protocol *ENcoder/DECoder* (ENDEC).

The features of the PrimeCell UART are covered under the following headings:

- *Features of the PrimeCell UART*
- *Programmable parameters* on page 1-3
- *Variations from the 16C550 UART* on page 1-4.

Note

Due to changes in the programmer's model, the PrimeCell UART (PL011) is not backwards compatible with the previous PrimeCell UART PL010.

1.1.1 Features of the PrimeCell UART

The PrimeCell UART provides:

- Compliance to the AMBA Specification (Rev 2.0) onwards for easy integration into SoC implementation.
- Programmable use of PrimeCell UART or IrDA SIR input/output.
- Separate 16x8 transmit and 16x12 receive *First-In, First-Out memory buffers* (FIFOs) to reduce CPU interrupts.
- Programmable FIFO disabling for 1-byte depth.
- Programmable baud rate generator. This enables division of the reference clock by (1x16) to (65535 x16) and generates an internal x16 clock. The divisor can be a fractional number enabling you to use any clock with a frequency >3.6864MHz as the reference clock.
- Standard asynchronous communication bits (start, stop and parity). These are added prior to transmission and removed on reception.
- Independent masking of transmit FIFO, receive FIFO, receive timeout, modem status, and error condition interrupts.
- Support for *Direct Memory Access* (DMA).

- False start bit detection.
- Line break generation and detection.
- Support of the modem control functions CTS, DCD, DSR, RTS, DTR, and RI.
- Programmable hardware flow control.
- Fully-programmable serial interface characteristics:
 - data can be 5, 6, 7, or 8 bits
 - even, odd, stick, or no-parity bit generation and detection
 - 1 or 2 stop bit generation
 - baud rate generation, dc up to UARTCLK_max_freq/16
- IrDA SIR ENDEC block providing:
 - programmable use of IrDA SIR or PrimeCell UART input/output
 - support of IrDA SIR ENDEC functions for data rates up to 115.2Kbits/second half-duplex
 - support of normal 3/16 and low-power (1.41–2.23µs) bit durations
 - programmable internal clock generator enabling division of reference clock by 1 to 256 for low-power mode bit duration.
- Identification registers that uniquely identify the PrimeCell UART. These can be used by an operating system to automatically configure itself.

1.1.2 Programmable parameters

The following key parameters are programmable:

- communication baud rate, integer, and fractional parts
- number of data bits
- number of stop bits
- parity mode
- FIFO enable (16 deep) or disable (1 deep)
- FIFO trigger levels selectable between 1/8, 1/4, 1/2, 3/4, and 7/8.
- internal nominal 1.8432MHz clock frequency (1.42–2.12MHz) to generate low-power mode shorter bit duration
- hardware flow control.

Additional test registers and modes are implemented for integration testing.

1.1.3 Variations from the 16C550 UART

The PrimeCell UART varies from the industry-standard 16C550 UART device as follows:

- receive FIFO trigger levels are 1/8, 1/4, 1/2, 3/4, and 7/8
- the internal register map address space, and the bit function of each register differ
- the deltas of the modem status signals are not available.

The following 16C550 UART features are not supported:

- 1.5 stop bits (1 or 2 stop bits only are supported)
- independent receive clock.

Chapter 2

Functional Overview

This chapter describes the major functional blocks of the ARM PrimeCell UART. It contains the following sections:

- *PrimeCell UART overview* on page 2-2
- *PrimeCell UART functional description* on page 2-4
- *IrDA SIR ENDEC functional description* on page 2-8
- *PrimeCell UART operation* on page 2-10
- *PrimeCell UART modem operation* on page 2-17
- *PrimeCell UART hardware flow control* on page 2-18
- *PrimeCell UART DMA interface* on page 2-20.

2.1 PrimeCell UART overview

The PrimeCell UART performs:

- serial-to-parallel conversion on data received from a peripheral device
- parallel-to-serial conversion on data transmitted to the peripheral device.

The CPU reads and writes data and control/status information through the AMBA APB interface. The transmit and receive paths are buffered with internal FIFO memories enabling up to 16-bytes to be stored independently in both transmit and receive modes.

The PrimeCell UART:

- includes a programmable baud rate generator that generates a common transmit and receive internal clock from the UART internal reference clock input, **UARTCLK**
- offers similar functionality to the industry-standard 16C550 UART device
- supports baud rates of up to 460.8Kbits/s, subject to **UARTCLK** reference clock frequency.

The PrimeCell UART operation and baud rate values are controlled by the line control register (UARTLCR_H) and the baud rate divisor registers (UARTIBRD and UARTFBRD).

The PrimeCell UART can generate:

- individually-maskable interrupts from the receive (including timeout), transmit, modem status and error conditions
- a single combined interrupt so that the output is asserted if any of the individual interrupts are asserted, and unmasked
- DMA request signals for interfacing with a *Direct Memory Access* (DMA) controller.

If a framing, parity, or break error occurs during reception, the appropriate error bit is set, and is stored in the FIFO. If an overrun condition occurs, the overrun register bit is set immediately and FIFO data is prevented from being overwritten.

You can program the FIFOs to be 1-byte deep providing a conventional double-buffered UART interface.

The modem status input signals *Clear To Send* (CTS), *Data Carrier Detect* (DCD), *Data Set Ready* (DSR), and *Ring Indicator* (RI) are supported. The output modem control lines, *Request To Send* (RTS), and *Data Terminal Ready* (DTR) are also supported.

There is a programmable hardware flow control feature that uses the **nUARTCTS** input and the **nUARTRTS** output to automatically control the serial data flow.

2.1.1 IrDA SIR block

The IrDA SIR block contains an IrDA SIR protocol ENDEC. The SIR protocol ENDEC can be enabled for serial communication through signals **nSIROUT** and **SIRIN** to an infrared transducer instead of using the UART signals **UARTTXD** and **UARTRXD**.

If the SIR protocol ENDEC is enabled, the **UARTTXD** line is held in the passive state (HIGH) and transitions of the modem status, or the **UARTRXD** line have no effect. The SIR protocol ENDEC can receive and transmit, but it is half-duplex only, so it cannot receive while transmitting, or transmit while receiving.

The IrDA SIR physical layer specifies a minimum 10ms delay between transmission and reception.

2.2 PrimeCell UART functional description

Figure 2-1 shows a block diagram of the PrimeCell UART.

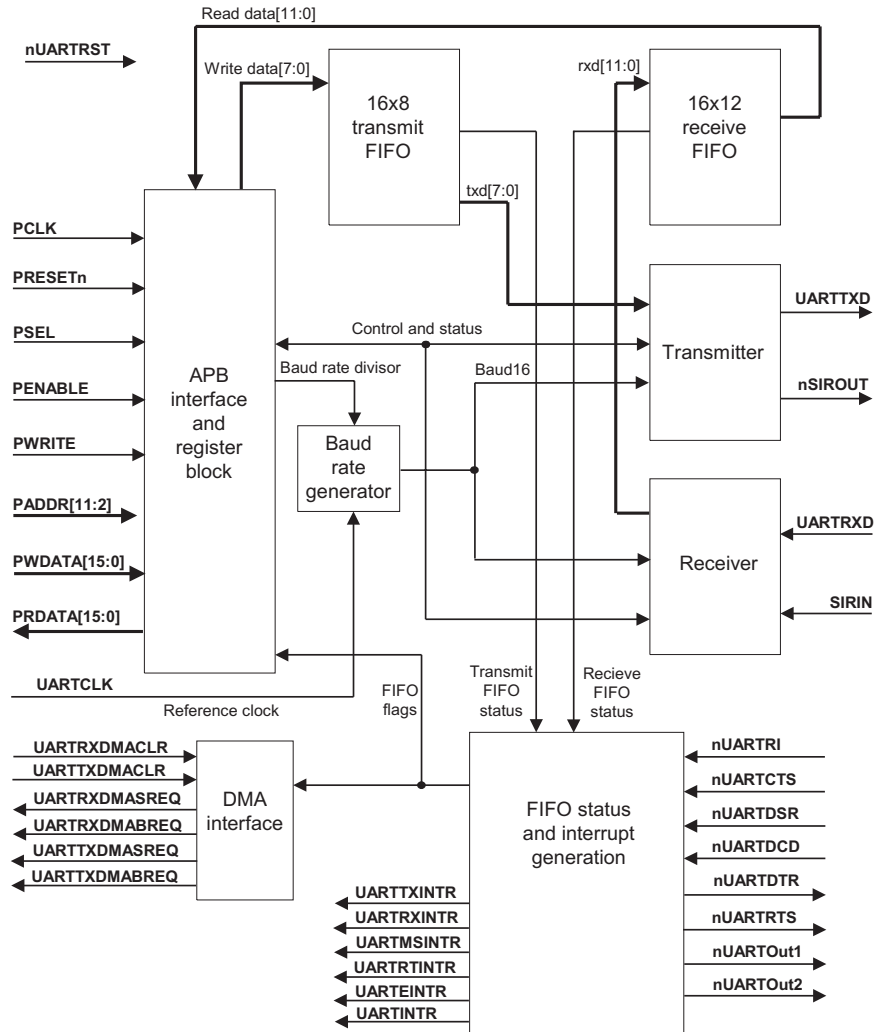


Figure 2-1 PrimeCell UART block diagram

Note

Test logic is not shown for clarity.

The functions of the PrimeCell UART are described in the following sections:

- *AMBA APB interface*
- *Register block*
- *Baud rate generator*
- *Transmit FIFO*
- *Receive FIFO* on page 2-6
- *Transmit logic* on page 2-6
- *Receive logic* on page 2-6
- *Interrupt generation logic* on page 2-6
- *DMA interface* on page 2-6
- *Synchronizing registers and logic* on page 2-7
- *Test registers and logic* on page 2-7.

2.2.1 AMBA APB interface

The AMBA APB interface generates read and write decodes for accesses to status/control registers and transmit/receive FIFO memories.

2.2.2 Register block

The register block stores data written, or to be read across the AMBA APB interface.

2.2.3 Baud rate generator

The baud rate generator contains free-running counters that generate the internal x16 clocks, **Baud16**, and the **IrLPBaud16** signal. **Baud16** provides timing information for UART transmit and receive control. **Baud16** is a stream of pulses with a width of one **UARTCLK** clock period and a frequency of 16 times the baud rate. **IrLPBaud16** provides timing information to generate the pulse width of the IrDA encoded transmit bit stream when in low-power mode.

2.2.4 Transmit FIFO

The transmit FIFO is an 8-bit wide, 16 location deep, FIFO memory buffer. CPU data written across the APB interface is stored in the FIFO until read out by the transmit logic. You can disable the transmit FIFO to act like a one-byte holding register.

2.2.5 Receive FIFO

The receive FIFO is a 12-bit wide, 16 location deep, FIFO memory buffer. Received data and corresponding error bits, are stored in the receive FIFO by the receive logic until read out by the CPU across the APB interface. The receive FIFO can be disabled to act like a one-byte holding register.

2.2.6 Transmit logic

The transmit logic performs parallel-to-serial conversion on the data read from the transmit FIFO. Control logic outputs the serial bit stream beginning with a start bit, data bits with the *Least Significant Bit* (LSB) first, followed by the parity bit, and then the stop bits according to the programmed configuration in control registers.

2.2.7 Receive logic

The receive logic performs serial-to-parallel conversion on the received bit stream after a valid start pulse has been detected. Overrun, parity, frame error checking, and line break detection are also performed, and their status accompanies the data that is written to the receive FIFO.

2.2.8 Interrupt generation logic

Individual maskable active HIGH interrupts are generated by the PrimeCell UART. A combined interrupt output is also generated as an OR function of the individual interrupt requests.

You can use the single combined interrupt with a system interrupt controller that provides another level of masking on a per-peripheral basis. This enables you to use modular device drivers that always know where to find the interrupt source control register bits.

You can also use the individual interrupt requests with a system interrupt controller that provides masking for the outputs of each peripheral. In this way, a global interrupt service routine can read the entire set of sources from one wide register in the system interrupt controller. This is attractive where the time to read from the peripheral registers is significant compared to the CPU clock speed in a real-time system.

The peripheral supports both the above methods.

2.2.9 DMA interface

The PrimeCell UART provides an interface to connect to the DMA controller. See *PrimeCell UART DMA interface* on page 2-20 for details.

2.2.10 Synchronizing registers and logic

The PrimeCell UART supports both asynchronous and synchronous operation of the clocks, **PCLK** and **UARTCLK**. Synchronization registers and handshaking logic have been implemented, and are active at all times. This has a minimal impact on performance or area. Synchronization of control signals is performed on both directions of data flow, that is from the **PCLK** to the **UARTCLK** domain, and from the **UARTCLK** to the **PCLK** domain.

2.2.11 Test registers and logic

There are registers and logic for functional block verification, and integration testing using TicTalk or code based vectors.

Test registers must not be read or written to during normal use.

The integration testing verifies that the UART has been wired into a system correctly. It enables each input and output to be both written to and read.

2.3 IrDA SIR ENDEC functional description

The IrDA SIR ENDEC comprises:

- *IrDA SIR transmit encoder*
- *IrDA SIR receive decoder* on page 2-9.

Figure 2-2 shows a block diagram of the IrDA SIR ENDEC.

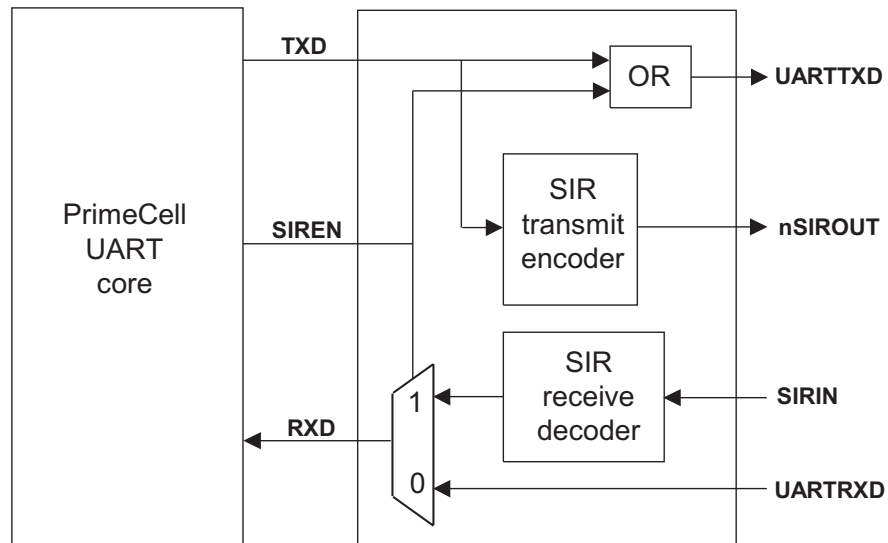


Figure 2-2 IrDA SIR ENDEC block diagram

2.3.1 IrDA SIR transmit encoder

The SIR transmit encoder modulates the *Non Return-to-Zero* (NRZ) transmit bit stream output from the PrimeCell UART. The IrDA SIR physical layer specifies use of a *Return To Zero, Inverted* (RZI) modulation scheme that represents logic 0 as an infrared light pulse. The modulated output pulse stream is transmitted to an external output driver and infrared *Light Emitting Diode* (LED).

In normal mode the transmitted pulse width is specified as three times the period of the internal x16 clock (**Baud16**), that is, $\frac{3}{16}$ of a bit period.

In low-power mode the transmit pulse width is specified as $\frac{3}{16}$ of a 115.2Kbits/s bit period. This is implemented as three times the period of a nominal 1.8432MHz clock (**IrLPBaud16**) derived from dividing down of **UARTCLK** clock. The frequency of **IrLPBaud16** is set up by writing the appropriate divisor value to **UARTILPR**.

The active low encoder output is normally LOW for the marking state (no light pulse). The encoder outputs a high pulse to generate an infrared light pulse representing a logic 0 or spacing state.

In normal and low power IrDA modes, when the fractional baud rate divider is used, the transmitted SIR pulse stream includes an increased amount of jitter. This jitter is because the Baud16 pulses cannot be generated at regular intervals when fractional division is used. That is, the Baud16 cycles have a different number of UARTCLK cycles. It can be shown that the worst case jitter in the SIR pulse stream can be up to three UARTCLK cycles.

This is within the limits of the SIR IrDA Specification where the maximum amount of jitter allowed is 13%, as long as the UARTCLK is > 3.6864 MHz and the maximum baud rate used for normal mode SIR is ≤ 115.2 kbps. Under these conditions, the jitter is less than 9%.

2.3.2 IrDA SIR receive decoder

The SIR receive decoder demodulates the return-to-zero bit stream from the infrared detector and outputs the received NRZ serial bit stream to the PrimeCell UART received data input. The decoder input is normally HIGH (marking state) in the idle state. The transmit encoder output has the opposite polarity to the decoder input.

A start bit is detected when the decoder input is LOW.

Regardless of being in normal or low-power mode, a start bit is deemed valid if the decoder is still LOW, one period of **IrLPBaud16** after the LOW was first detected. This enables a normal-mode UART to receive data from a low-power mode UART, that can transmit pulses as small as $1.41\mu\text{s}$.

2.4 PrimeCell UART operation

The operation of the PrimeCell UART is described in the following sections:

- *Interface reset*
- *Clock signals*
- *PrimeCell UART operation on page 2-11*
- *IrDA SIR operation on page 2-14*
- *PrimeCell UART character frame on page 2-15*
- *IrDA data modulation on page 2-15.*

2.4.1 Interface reset

The PrimeCell UART and IrDA SIR ENDEC are reset by the global reset signal **PRESETn** and a block-specific reset signal **nUARTRST**. An external reset controller must use **PRESETn** to assert **nUARTRST** asynchronously and negate it synchronously to **UARTCLK**. **PRESETn** must be asserted LOW for a period long enough to reset the slowest block in the on-chip system, and then be taken HIGH again. The PrimeCell UART requires **PRESETn** to be asserted LOW for at least one period of **PCLK**.

The values of the registers after reset are detailed in Chapter 3 *Programmer's Model*.

2.4.2 Clock signals

The frequency selected for **UARTCLK** must accommodate the desired range of baud rates:

$$F_{\text{UARTCLK}}(\text{min}) \geq 16 \times \text{baud_rate}(\text{max})$$

$$F_{\text{UARTCLK}}(\text{max}) \leq 16 \times 65535 \times \text{baud_rate}(\text{min})$$

For example, for a range of baud rates from 110 baud to 460800 baud the **UARTCLK** frequency must be within the range 7.3728MHz to 115MHz.

The frequency of **UARTCLK** must also be within the required error limits for all baud rates to be used.

There is also a constraint on the ratio of clock frequencies for **PCLK** to **UARTCLK**. The frequency of **UARTCLK** must be no more than $5/3$ times faster than the frequency of **PCLK**:

$$F_{\text{UARTCLK}} \leq \frac{5}{3} \times F_{\text{PCLK}}$$

This allows sufficient time to write the received data to the receive FIFO.

2.4.3 PrimeCell UART operation

Control data is written to the PrimeCell UART line control register, `UARTLCR_H`. This register is 29 bits wide internally, but is externally accessed through the AMBA APB bus by three writes to register locations, `UARTLCR_H`, `UARTIBRD`, and `UARTFBRD`. `UARTLCR_H` defines:

- transmission parameters
- word length
- buffer mode
- number of transmitted stop bits
- parity mode
- break generation.

`UARTIBRD` and `UARTFBRD` together define the baud rate divisor.

Fractional baud rate divider

The baud rate divisor is a 22-bit number consisting of a 16-bit integer and a 6-bit fractional part. This is used by the baud rate generator to determine the bit period. The fractional baud rate divider enables the use of any clock with a frequency $>3.6864\text{MHz}$ to act as `UARTCLK`, while it is still possible to generate all the standard baud rates.

The 16-bit integer is loaded through the `UARTIBRD` register. The 6-bit fractional part is loaded into the `UARTFBRD` register. The Baud Rate Divisor has the following relationship to `UARTCLK`:

$$\text{Baud Rate Divisor} = \text{UARTCLK} / (16 \times \text{Baud Rate}) = \text{BRD}_I + \text{BRD}_F$$

where BRD_I is the integer part and BRD_F is the fractional part separated by a decimal point as shown in Figure 2-3.

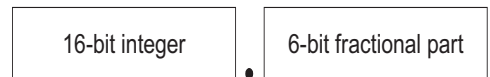


Figure 2-3 Baud rate divisor

You can calculate the 6-bit number (m) by taking the fractional part of the required baud rate divisor and multiplying it by 64 (that is, 2^n , where n is the width of the `UARTFBRD` register) and adding 0.5 to account for rounding errors:

$$m = \text{integer}(\text{BRD}_F * 2^n + 0.5)$$

See Example 3-1 on page 3-11.

An internal clock enable signal, **Baud16**, is generated, and is a stream of one **UARTCLK** wide pulses with an average frequency of 16 times the desired baud rate. This signal is then divided by 16 to give the transmit clock. A low number in the baud rate divisor gives a short bit period, and a high number in the baud rate divisor gives a long bit period.

Data transmission or reception

Data received or transmitted is stored in two 16-byte FIFOs, though the receive FIFO has an extra four bits per character for status information.

For transmission, data is written into the transmit FIFO. If the PrimeCell UART is enabled, it causes a data frame to start transmitting with the parameters indicated in **UARTLCR_H**. Data continues to be transmitted until there is no data left in the transmit FIFO. The **BUSY** signal goes HIGH as soon as data is written to the transmit FIFO (that is, the FIFO is non-empty) and remains asserted HIGH while data is being transmitted. **BUSY** is negated only when the transmit FIFO is empty, and the last character has been transmitted from the shift register, including the stop bits. **BUSY** can be asserted HIGH even though the PrimeCell UART might no longer be enabled.

For each sample of data, three readings are taken and the majority value is kept. In the following paragraphs the middle sampling point is defined, and one sample is taken either side of it.

When the receiver is idle (**UARTRXD** continuously 1, in the marking state) and a LOW is detected on the data input (a start bit has been received), the receive counter, with the clock enabled by **Baud16**, begins running and data is sampled on the eighth cycle of that counter in normal UART mode, or the fourth cycle of the counter in SIR mode to allow for the shorter logic 0 pulses (half way through a bit period).

The start bit is valid if **UARTRXD** is still LOW on the eighth cycle of **Baud16**, otherwise a false start bit is detected and it is ignored.

If the start bit was valid, successive data bits are sampled on every 16th cycle of **Baud16** (that is, one bit period later) according to the programmed length of the data characters. The parity bit is then checked if parity mode was enabled.

Lastly, a valid stop bit is confirmed if **UARTRXD** is HIGH, otherwise a framing error has occurred. When a full word is received, the data is stored in the receive FIFO, with any error bits associated with that word (see Table 2-1 on page 2-13).

Error bits

Three error bits are stored in bits [10:8] of the receive FIFO, and are associated with a particular character. There is an additional error that indicates an overrun error and this is stored in bit 11 of the receive FIFO.

Overrun bit

The overrun bit is not associated with the character in the receive FIFO. The overrun error is set when the FIFO is full, and the next character is completely received in the shift register. The data in the shift register is overwritten, but it is not written into the FIFO. When an empty location is available in the receive FIFO, and another character is received, the state of the overrun bit is copied into the receive FIFO along with the received character. The overrun state is then cleared. Table 2-1 shows the bit functions of the receive FIFO.

Table 2-1 Receive FIFO bit functions

FIFO bit	Function
11	Overrun indicator
10	Break error
9	Parity error
8	Framing error
7:0	Received data

Disabling the FIFOs

Additionally, you can disable the FIFOs. In this case, the transmit and receive sides of the PrimeCell UART have 1-byte holding registers (the bottom entry of the FIFOs). The overrun bit is set when a word has been received, and the previous one was not yet read. In this implementation, the FIFOs are not physically disabled, but the flags are manipulated to give the illusion of a 1-byte register.

System and diagnostic loopback testing

You can perform loopback testing for PrimeCell UART data by setting the *Loop Back Enable* (LBE) bit to 1 in the control register UARTCR (bit 7).

Data transmitted on **UARTTXD** is received on the **UARTRXD** input.

2.4.4 IrDA SIR operation

The IrDA SIR ENDEC provides functionality that converts between an asynchronous PrimeCell UART data stream, and half-duplex serial SIR interface. No analog processing is performed on-chip. The role of the SIR ENDEC is to provide a digital encoded output, and decoded input to the PrimeCell UART. There are two modes of operation:

- In normal **IrDA** mode, a zero logic level is transmitted as high pulse of $3/16$ th duration of the selected baud rate bit period on the **nSIROUT** signal, while logic one levels are transmitted as a static LOW signal. These levels control the driver of an infrared transmitter, sending a pulse of light for each zero. On the reception side, the incoming light pulses energize the photo transistor base of the receiver, pulling its output LOW. This drives the **SIRIN** signal LOW.
- In low-power **IrDA** mode, the width of the transmitted infrared pulse is set to three times the period of the internally generated **IrLPBaud16** signal (1.63 μ s, assuming a nominal 1.8432MHz frequency) by changing the appropriate bit in **UARTCR**.

In both normal and low-power **IrDA** modes:

- during transmission, the PrimeCell UART data bit is used as the base for encoding
- during reception, the decoded bits are transferred to the PrimeCell UART receive logic.

The IrDA SIR physical layer specifies a half-duplex communication link, with a minimum 10ms delay between transmission and reception. This delay must be generated by software because it is not supported by the PrimeCell UART. The delay is required because the Infrared receiver electronics might become biased, or even saturated from the optical power coupled from the adjacent transmitter LED. This delay is known as latency, or receiver setup time.

The **IrLPBaud16** signal is generated by dividing down the **UARTCLK** signal according to the low-power divisor value written to **UARTILPR**.

The low-power divisor value is calculated as:

$$\text{Low-power divisor} = (\mathbf{F_{UARTCLK}} / \mathbf{F_{IrLPBaud16}})$$

where **F_{IrLPBaud16}** is nominally 1.8432MHz.

The divisor must be chosen so that $1.42\text{MHz} < \mathbf{F_{IrLPBaud16}} < 2.12\text{MHz}$.

System and diagnostic loopback testing

It is possible to perform loopback testing for SIR data by:

- Setting the *Loop Back Enable* (LBE) bit to 1 in the control register UARTCR (bit 7).
- Setting the SIRTEST bit to 1 in the test register UARTTCR (bit 2).

Data transmitted on **nSIROUT** is received on the SIRIN input.

Note

This is the only occasion that a test register needs to be accessed during normal operation.

2.4.5 PrimeCell UART character frame

The PrimeCell UART character frame is shown in Figure 2-4.

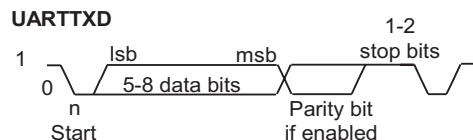


Figure 2-4 PrimeCell UART character frame

2.4.6 IrDA data modulation

The effect of IrDA $3/16$ data modulation can be seen in Figure 2-5 on page 2-16.

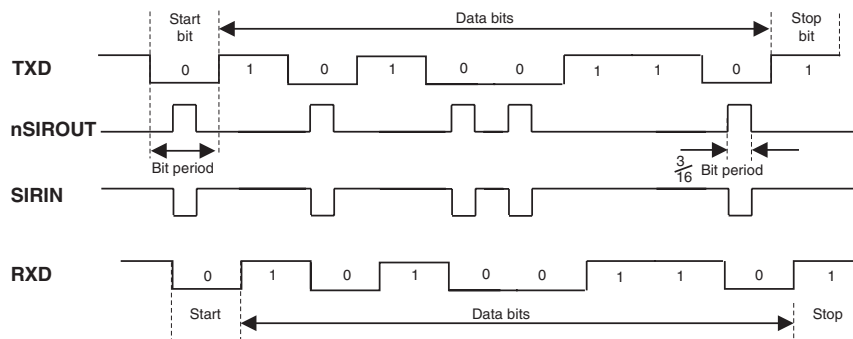


Figure 2-5 IrDA data modulation (3/16)

2.5 PrimeCell UART modem operation

You can use the PrimeCell UART to support both the *Data Terminal Equipment* (DTE) and *Data Communication Equipment* (DCE) modes of operation. Figure 2-1 on page 2-4 shows the modem signals in the DTE mode. For DCE mode, Table 2-2 shows the meaning of the signals.

Table 2-2 Meaning of modem input/output in DTE and DCE modes

Port Name	Meaning	
	DTE	DCE
nUARTCTS	Clear to send	Request to send
nUARTDSR	Data set ready	Data terminal ready
nUARTDCD	Data carrier detect	-
nUARTRI	Ring indicator	-
nUARTRTS	Request to send	Clear to send
nUARTDTR	Data terminal ready	Data set ready
nUARTOUT1	-	Data carrier detect
nUARTOUT2	-	Ring indicator

2.6 PrimeCell UART hardware flow control

The hardware flow control feature is fully selectable, and enables you to control the serial data flow by using the **nUARTRTS** output and **nUARTCTS** input signals. Figure 2-6 shows how two devices can communicate with each other using hardware flow control.

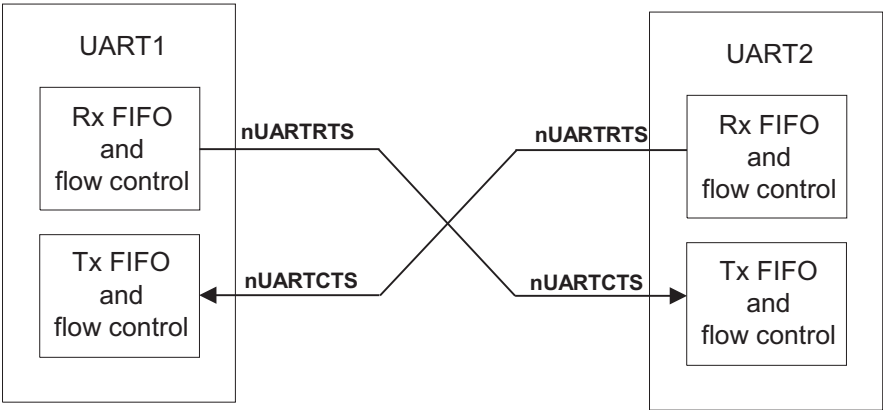


Figure 2-6 Hardware flow control between two similar devices

When the RTS flow control is enabled, the **nUARTRTS** signal is asserted until the receive FIFO is filled up to the programmed watermark level. When the CTS flow control is enabled, the transmitter can only transmit data when the **nUARTCTS** signal is asserted.

The hardware flow control is selectable through bits 14 (**RTSEn**) and 15 (**CTSEn**) in the UART control register (UARTCR). Table 2-3 shows how you must set the bits to enable RTS and CTS flow control both simultaneously, and independently.

Table 2-3 Control bits to enable and disable hardware flow control

CTSEn bit 15 in UARTCR	RTSEn bit 14 in UARTCR	Description
1	1	Both RTS and CTS flow control enabled

Table 2-3 Control bits to enable and disable hardware flow control (continued)

CTSEn bit 15 in UARTCR	RTSEn bit 14 in UARTCR	Description
1	0	Only CTS flow control enabled
0	1	Only RTS flow control enabled
0	0	Both RTS and CTS flow control disabled

Note

When RTS flow control is enabled, the software cannot control the **nUARTRTS** line through bit 11 of the UART control register.

2.6.1 **RTS flow control**

The RTS flow control logic is linked to the programmable receive FIFO watermark levels. When RTS flow control is enabled, the **nUARTRTS** is asserted until the receive FIFO is filled up to the watermark level. When the receive FIFO watermark level is reached, the **nUARTRTS** signal is deasserted, indicating that there is no more room to receive any more data. The transmission of data is expected to cease after the current character has been transmitted.

The **nUARTRTS** signal is reasserted when data has been read out of the receive FIFO so that it is filled to less than the watermark level. If RTS flow control is disabled and the UART is still enabled, then data is received until the receive FIFO is full, or no more data is transmitted to it.

2.6.2 **CTS flow control**

If CTS flow control is enabled, then the transmitter checks the **nUARTCTS** signal before transmitting the next byte. If the **nUARTCTS** signal is asserted, it transmits the byte otherwise transmission does not occur.

The data continues to be transmitted while **nUARTCTS** is asserted, and the transmit FIFO is not empty. If the transmit FIFO is empty and the **nUARTCTS** signal is asserted no data is transmitted.

If the **nUARTCTS** signal is deasserted and CTS flow control is enabled, then the current character transmission is completed before stopping. If CTS flow control is disabled and the UART is enabled, then the data continues to be transmitted until the transmit FIFO is empty.

2.7 PrimeCell UART DMA interface

The PrimeCell UART provides an interface to connect to the DMA controller. The DMA operation of the UART is controlled through the UART DMA control register, **UARTDMACR**. The DMA interface includes the following signals:

For receive:

UARTRXDMASREQ

Single character DMA transfer request, asserted by the UART. For receive, one character consists of up to 12 bits. This signal is asserted when the receive FIFO contains at least one character.

UARTRXDMABREQ

Burst DMA transfer request, asserted by the UART. This signal is asserted when the receive FIFO contains more characters than the programmed watermark level. You can program the watermark level for each FIFO through the **UARTIFLS** register.

UARTRXDMACLR

DMA request clear, asserted by the DMA controller to clear the receive request signals. If DMA burst transfer is requested, the clear signal is asserted during the transfer of the last data in the burst.

For transmit:

UARTTXDMASREQ

Single character DMA transfer request, asserted by the UART. For transmit one character consists of up to eight bits. This signal is asserted when there is at least one empty location in the transmit FIFO.

UARTTXDMABREQ

Burst DMA transfer request, asserted by the UART. This signal is asserted when the transmit FIFO contains less characters than the watermark level. You can program the watermark level for each FIFO through the **UARTIFLS** register.

UARTTXDMACLR

DMA request clear, asserted by the DMA controller to clear the transmit request signals. If DMA burst transfer is requested, the clear signal is asserted during the transfer of the last data in the burst.

The burst transfer and single transfer request signals are not mutually exclusive, they can both be asserted at the same time. For example, when there is more data than the watermark level in the receive FIFO, the burst transfer request and the single transfer

request are asserted. When the amount of data left in the receive FIFO is less than the watermark level, the single request only is asserted. This is useful for situations where the number of characters left to be received in the stream is less than a burst.

For example, say 19 characters have to be received and the watermark level is programmed to be four. The DMA controller then transfers four bursts of four characters and three single transfers to complete the stream.

Note

For the remaining three characters the UART cannot assert the burst request.

Each request signal remains asserted until the relevant **DMACLR** signal is asserted. After the request clear signal is deasserted, a request signal can become active again, depending on the conditions described above. All request signals are deasserted if the UART is disabled or the DMA enable signal is cleared.

When the UART is in the FIFO disabled mode, only the DMA single transfer mode can operate, since only one character can be transferred to, or from the FIFOs at any time. **UARTRXDMASREQ** and **UARTTXDMASREQ** are the only request signals that can be asserted. When the UART is in the FIFO enabled mode, data transfers can be made by either single or burst transfers depending on the programmed watermark level and the amount of data in the FIFO. Table 2-4 shows the trigger points for **DMABREQ** depending on the watermark level, for both the transmit and receive FIFOs.

Table 2-4 DMA trigger points for the transmit and receive FIFOs

Watermark level	Burst length	
	Transmit (number of empty locations)	Receive (number of filled locations)
1/8	14	2
1/4	12	4
1/2	8	8
3/4	4	12
7/8	2	14

In addition to the above, the DMAONERR bit in the DMA control register supports the use of the receive error interrupt, **UARTEINTR**. It enables the DMA receive request outputs, **UARTRXDMASREQ** or **UARTRXDMABREQ**, to be masked out when the

UART error interrupt, **UARTEINTR**, is asserted. The DMA receive request outputs remain inactive until the **UARTEINTR** is cleared. The DMA transmit request outputs are unaffected.

Figure 2-7 shows the timing diagram for both a single transfer request and a burst transfer request with the appropriate DMA clear signal. The signals are all synchronous to **PCLK**. For the sake of clarity it is assumed that there is no synchronization of the request signals in the DMA controller.

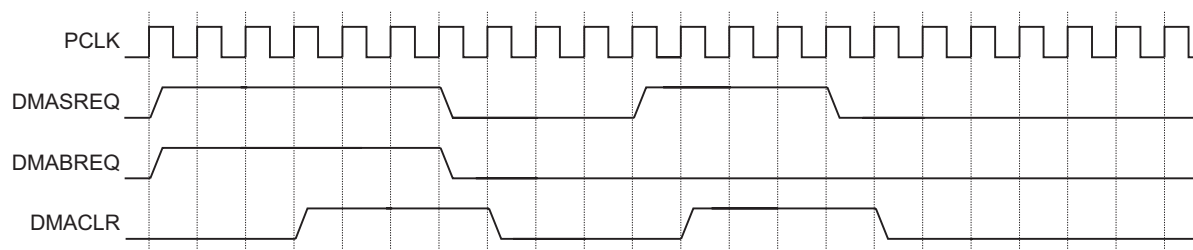


Figure 2-7 DMA transfer waveforms

Chapter 3

Programmer's Model

This chapter describes the ARM PrimeCell UART (PL011) registers and provides details needed when programming the microcontroller. It contains the following sections:

- *About the programmer's model* on page 3-2
- *Summary of PrimeCell UART registers* on page 3-3
- *Register descriptions* on page 3-5
- *Interrupts* on page 3-29.

3.1 About the programmer's model

The base address of the PrimeCell UART is not fixed, and can be different for any particular system implementation. However, the offset of any particular register from the base address is fixed.

The following locations are reserved, and must not be used during normal operation:

- locations at offsets 0x008 through 0x014, 0x01C are reserved and must not be accessed
- locations at offsets 0x04C through 0x07C are reserved for possible future extensions
- locations at offsets 0x080 through 0x08C are reserved for test purposes
- locations at offsets 0x90 through 0xFCC are reserved for future test purposes
- location at offsets 0xFD0 through 0xFDC are used for future identification registers
- location at offsets 0xFE0 through 0xFFC are used for identification registers.

3.2 Summary of PrimeCell UART registers

The PrimeCell UART registers are shown in Table 3-1.

Table 3-1 PrimeCell UART register summary

Address	Type	Width	Reset value	Name	Description
UART base + 0x000	Read/write	12/8	0x---	UARTDR	Data read or written from the interface. It is 12 bits wide on a read, and 8 on a write.
UART base + 0x004	Read/write	4/0	0x0	UARTRSR/ UARTECR	Receive status register (read)/ error clear register (write).
UART base + 0x008-0x014	-	-	-	-	Reserved.
UART base + 0x018	Read	9	0b-10010---	UARTFR	Flag register (read only).
UART base + 0x01C	-	-	-	-	Reserved.
UART base + 0x020	Read/write	8	0x00	UARTILPR	IrDA low-power counter register.
UART base + 0x024	Read/write	16	0x0000	UARTIBRD	Integer baud rate divisor register.
UART base + 0x028	Read/write	6	0x00	UARTFBRD	Fractional baud rate divisor register.
UART base + 0x02C	Read/write	8	0x00	UARTLCR_H	Line control register, HIGH byte.
UART base + 0x030	Read/write	16	0x0300	UARTCR	Control register.
UART base + 0x034	Read/write	6	0x12	UARTIFLS	Interrupt FIFO level select register.
UART base + 0x038	Read/write	11	0x000	UARTIMSC	Interrupt mask set/clear.
UART base + 0x03C	Read	11	0x00-	UARTRIS	Raw interrupt status.

Table 3-1 PrimeCell UART register summary (continued)

Address	Type	Width	Reset value	Name	Description
UART base + 0x040	Read	11	0x00-	UARTMIS	Masked interrupt status.
UART base + 0x044	Write	11	-	UARTICR	Interrupt clear register.
UART base + 0x048	Read/write	3	0x00	UARTDMACR	DMA control register.
UART base + 0x04C-07C	-	-	-	-	Reserved.
UART base + 0x080-0x08C	-	-	-	-	Reserved (for test purposes).
UART base + 0x090-FCC	-	-	-	-	Reserved.
UART base + 0xFD0-FDC	-	-	-	-	Reserved for future ID expansion.
UART base + 0xFE0	Read	8	0x11	UARTPeriphID0	Peripheral identification register bits [7:0].
UART base + 0xFE4	Read	8	0x10	UARTPeriphID1	Peripheral identification register bits [15:8].
UART base + 0xFE8	Read	8	0x14	UARTPeriphID2	Peripheral identification register bits [23:16].
UART base + 0xFEC	Read	8	0x00	UARTPeriphID3	Peripheral identification register bits [31:24].
UART base + 0xFF0	Read	8	0x0D	UARTPCellID0	PrimeCell identification register bits [7:0].
UART base + 0xFF4	Read	8	0xF0	UARTPCellID1	PrimeCell identification register bits [15:8].
UART base + 0xFF8	Read	8	0x05	UARTPCellID2	PrimeCell identification register bits [23:16].
UART base + 0xFFC	Read	8	0xB1	UARTPCellID3	PrimeCell identification register bits [31:24].

3.3 Register descriptions

The following PrimeCell UART registers are described in this section:

- *Data register, UARTDR*
- *Receive status register/error clear register, UARTRSR/UARTECR on page 3-7*
- *Flag register, UARTFR on page 3-8*
- *IrDA low-power counter register, UARTILPR on page 3-9*
- *Integer baud rate register, UARTIBRD on page 3-10*
- *Fractional baud rate register, UARTFBRD on page 3-10*
- *Line control register, UARTLCR_H on page 3-12*
- *Control register, UARTCR on page 3-15*
- *Interrupt FIFO level select register, UARTIFLS on page 3-17*
- *Interrupt mask set/clear register, UARTIMSC on page 3-18*
- *Raw interrupt status register, UARTRIS on page 3-19*
- *Masked interrupt status register, UARTMIS on page 3-21*
- *Interrupt clear register, UARTICR on page 3-22*
- *DMA control register, UARTDMACR on page 3-23*
- *Peripheral identification registers, UARTPeriphID0-3 on page 3-23*
- *PrimeCell identification registers, UARTPCellID0-3 on page 3-26.*

3.3.1 Data register, UARTDR

The UARTDR register is the data register.

For words to be transmitted:

- if the FIFOs are enabled, data written to this location is pushed onto the transmit FIFO
- if the FIFOs are not enabled, data is stored in the transmitter holding register (the bottom word of the transmit FIFO).

The write operation initiates transmission from the PrimeCell UART. The data is prefixed with a start bit, appended with the appropriate parity bit (if parity is enabled), and a stop bit. The resultant word is then transmitted.

For received words:

- if the FIFOs are enabled, the data byte and the 4-bit status (break, frame, parity, and overrun) is pushed onto the 12-bit wide receive FIFO
- if the FIFOs are not enabled, the data byte and status are stored in the receiving holding register (the bottom word of the receive FIFO).

The received data byte is read by performing reads from the UARTDR register along with the corresponding status information. The status information can also be read by a read of the UARTRSR/UARTECR register as shown in Table 3-2.

Table 3-2 UARTDR register

Bits	Name	Type	Function
15:12	RES	-	Reserved.
11	Overrun Error (OE)	Read	This bit is set to 1 if data is received and the receive FIFO is already full. This is cleared to 0 once there is an empty space in the FIFO and a new character can be written to it.
10	Break Error (BE)	Read	This bit is set to 1 if a break condition was detected, indicating that the received data input was held LOW for longer than a full-word transmission time (defined as start, data, parity and stop bits). In FIFO mode, this error is associated with the character at the top of the FIFO. When a break occurs, only one 0 character is loaded into the FIFO. The next character is only enabled after the receive data input goes to a 1 (marking state), and the next valid start bit is received.
9	Parity Error (PE)	Read	When this bit is set to 1, it indicates that the parity of the received data character does not match the parity selected as defined by bits 2 and 7 of the UARTLCR_H register. In FIFO mode, this error is associated with the character at the top of the FIFO.
8	Framing Error (FE)	Read	When this bit is set to 1, it indicates that the received character did not have a valid stop bit (a valid stop bit is 1). In FIFO mode, this error is associated with the character at the top of the FIFO.
7:0	DATA	Read/ write	Receive (read) data character. Transmit (write) data character.

———— **Note** ————

You must disable the PrimeCell UART before any of the control registers are reprogrammed.

When the PrimeCell UART is disabled in the middle of transmission or reception, it completes the current character before stopping.

3.3.2 Receive status register/error clear register, UARTRSR/UARTECR

The UARTRSR/UARTECR register is the receive status register/error clear register.

Receive status can also be read from UARTRSR. If the status is read from this register, then the status information for break, framing and parity corresponds to the data character read from UARTDR prior to reading UARTRSR. The status information for overrun is set immediately when an overrun condition occurs.

A write to UARTECR clears the framing, parity, break, and overrun errors. All the bits are cleared to 0 on reset. Table 3-3 shows the bit assignment of the UARTRSR/UARTECR register.

Table 3-3 UARTRSR/UARTECR register

Bits	Name	Type	Function
7:0	-	Write	A write to this register clears the framing, parity, break, and overrun errors. The data value is not important.
7:4	-	Read	Reserved, unpredictable when read.
3	Overrun Error (OE)	Read	This bit is set to 1 if data is received and the FIFO is already full. This bit is cleared to 0 by a write to UARTECR. The FIFO contents remain valid since no further data is written when the FIFO is full, only the contents of the shift register are overwritten. The CPU must now read the data in order to empty the FIFO.
2	Break Error (BE)	Read	This bit is set to 1 if a break condition was detected, indicating that the received data input was held LOW for longer than a full-word transmission time (defined as start, data, parity, and stop bits). This bit is cleared to 0 after a write to UARTECR. In FIFO mode, this error is associated with the character at the top of the FIFO. When a break occurs, only one 0 character is loaded into the FIFO. The next character is only enabled after the receive data input goes to a 1 (marking state) and the next valid start bit is received.
1	Parity Error (PE)	Read	When this bit is set to 1, it indicates that the parity of the received data character does not match the parity selected as defined by bits 2 and 7 of the UARTRSR register. This bit is cleared to 0 by a write to UARTECR. In FIFO mode, this error is associated with the character at the top of the FIFO.
0	Framing Error (FE)	Read	When this bit is set to 1, it indicates that the received character did not have a valid stop bit (a valid stop bit is 1). This bit is cleared to 0 by a write to UARTECR. In FIFO mode, this error is associated with the character at the top of the FIFO.

Note

The received data character must be read first from UARTDR before reading the error status associated with that data character from UARTRSR. This read sequence cannot be reversed, because the status register UARTRSR is updated only when a read occurs from the data register UARTDR. However, the status information can also be obtained by reading the UARTDR register.

3.3.3 Flag register, UARTFR

The UARTFR register is the flag register. After reset TXFF, RXFF, and BUSY are 0, and TXFE and RXFE are 1. Table 3-4 shows the bit assignment of the UARTFR register.

Table 3-4 UARTFR register

Bits	Name	Type	Function
15:9	-	-	Reserved, do not modify, read as zero.
8	Ring Indicator (RI)	Read	This bit is the complement of the PrimeCell UART ring indicator (nUARTRI) modem status input. That is, the bit is 1 when the modem status input is 0.
7	Transmit FIFO Empty (TXFE)	Read	The meaning of this bit depends on the state of the FEN bit in the UARTLCR_H register. If the FIFO is disabled, this bit is set when the transmit holding register is empty. If the FIFO is enabled, the TXFE bit is set when the transmit FIFO is empty.
6	Receive FIFO Full (RXFF)	Read	The meaning of this bit depends on the state of the FEN bit in the UARTLCR_H register. If the FIFO is disabled, this bit is set when the receive holding register is full. If the FIFO is enabled, the RXFF bit is set when the receive FIFO is full.
5	Transmit FIFO Full (TXFF)	Read	The meaning of this bit depends on the state of the FEN bit in the UARTLCR_H register. If the FIFO is disabled, this bit is set when the transmit holding register is full. If the FIFO is enabled, the TXFF bit is set when the transmit FIFO is full.
4	Receive FIFO Empty (RXFE)	Read	The meaning of this bit depends on the state of the FEN bit in the UARTLCR_H register. If the FIFO is disabled, this bit is set when the receive holding register is empty. If the FIFO is enabled, the RXFE bit is set when the receive FIFO is empty.

Table 3-4 UARTFR register (continued)

Bits	Name	Type	Function
3	UART Busy (BUSY)	Read	If this bit is set to 1, the PrimeCell UART is busy transmitting data. This bit remains set until the complete byte, including all the stop bits, has been sent from the shift register. This bit is set as soon as the transmit FIFO becomes non-empty (regardless of whether the PrimeCell UART is enabled or not).
2	Data Carrier Detect (DCD)	Read	This bit is the complement of the PrimeCell UART data carrier detect (nUARTDCD) modem status input. That is, the bit is 1 when the modem status input is 0.
1	Data Set Ready (DSR)	Read	This bit is the complement of the PrimeCell UART data set ready (nUARTDSR) modem status input. That is, the bit is 1 when the modem status input is 0.
0	Clear To Send (CTS)	Read	This bit is the complement of the PrimeCell UART clear to send (nUARTCTS) modem status input. That is, the bit is 1 when the modem status input is 0.

3.3.4 IrDA low-power counter register, UARTILPR

The UARTILPR register is the IrDA low-power counter register. This is an 8-bit read/write register that stores the low-power counter divisor value used to generate the **IrLPBaud16** signal by dividing down of **UARTCLK**. All the bits are cleared to 0 when reset. Table 3-5 shows the bit assignment of the UARTILPR register.

Table 3-5 UARTILPR register

Bits	Name	Type	Function
7:0	IrDA Low Power Divisor [7:0] (ILPDVSR)	Read/write	8-bit low-power divisor value. These bits are cleared to 0 at reset.

The **IrLPBaud16** signal is generated by dividing down the **UARTCLK** signal according to the low-power divisor value written to UARTILPR.

The low-power divisor value is calculated as follows:

$$\text{low-power divisor (ILPDVSR)} = (\mathbf{F_{UARTCLK}} / \mathbf{F_{IrLPBaud16}})$$

where **F_{IrLPBaud16}** is nominally 1.8432MHz.

You must chose the divisor so that $1.42\text{MHz} < \mathbf{F_{IrLPBaud16}} < 2.12\text{MHz}$, that results in a low-power pulse duration of 1.41–2.11µs (three times the period of **IrLPBaud16**).

The minimum frequency of **IrLPBaud16** ensures that pulses less than one period of **IrLPBaud16** are rejected, but that pulses greater than 1.4µs are accepted as valid pulses.

———— **Note** —————

Zero is an illegal value. Programming a zero value results in no **IrLPBaud16** pulses being generated.

3.3.5 Integer baud rate register, UARTIBRD

The UARTIBRD register is the integer part of the baud rate divisor value. All the bits are cleared to 0 on reset. Table 3-6 shows the bit assignment of the UARTIBRD register.

Table 3-6 UARTIBRD register

Bits	Name	Type	Function
15:0	Baud Rate Integer[15:0] (BAUD DIVINT)	Read/write	The integer baud rate divisor. These bits are cleared to 0 on reset.

3.3.6 Fractional baud rate register, UARTFBRD

The UARTFBRD register is the fractional part of the baud rate divisor value. All the bits are cleared to 0 on reset. Table 3-7 shows the bit assignment of register of the UARTFBRD register.

Table 3-7 UARTFBRD register

Bits	Name	Type	Function
5:0	Baud Rate fraction[5:0] (BAUD DIVFRAC)	Read/write	The fractional baud rate divisor. These bits are cleared to 0 on reset.

The baud rate divisor is calculated as follows:

Baud rate divisor BAUDDIV = (**F**_{UARTCLK}/ { 16 * Baud rate })

where **F**_{UARTCLK} is the UART reference clock frequency.

The BAUDDIV is comprised of the integer value (BAUD DIVINT) and the fractional value (BAUD DIVFRAC).

Note

The contents of the UARTIBRD and UARTFBRD registers are not updated until transmission or reception of the current character is complete.

The minimum divide ratio possible is 1 and the maximum is 65535(2¹⁶ - 1). That is, UARTIBRD = 0 is invalid and UARTFBRD is ignored when this is the case.

Similarly, when UARTIBRD = 65535 (that is 0xFFFF), then UARTFBRD must not be greater than zero. If this is exceeded it results in an aborted transmission or reception.

Example 3-1 is an example of how to calculate the divisor value.

Example 3-1 Calculating the divisor value

If the required baud rate is 230400 and **UARTCLK** = 4MHz then:

Baud Rate Divisor = $(4 * 10^6)/(16 * 230400) = 1.085$

Therefore, BRD_I = 1 and BRD_F = 0.085,

Therefore, fractional part, m = integer((0.085 * 64) + 0.5) = 5

Generated baud rate divider = $1 + 5/64 = 1.078$

Generated baud rate = $(4 * 10^6)/(16 * 1.078) = 231911$

Error = $(231911 - 230400)/230400 * 100 = 0.656\%$

The maximum error using a 6-bit UARTFBRD register = $1/64 * 100 = 1.56\%$. This occurs when m = 1, and the error is cumulative over 64 clock ticks.

Table 3-8 shows some typical bit rates and their corresponding divisors, given the PrimeCell UART clock frequency of 7.3728MHz. These values do not use the fractional divider so the value in the UARTFBRD register is zero.

Table 3-8 Typical baud rates and divisors

Programmed integer divisor	Bit rate (bps)
0x1	460800
0x2	230400
0x4	115200

Table 3-8 Typical baud rates and divisors (continued)

Programmed integer divisor	Bit rate (bps)
0x6	76800
0x8	57600
0xC	38400
0x18	19200
0x20	14400
0x30	9600
0xC0	2400
0x180	1200
0x105D	110

Table 3-9 shows some required bit rates and their corresponding integer and fractional divisor values and generated bit rates given a clock frequency of 4MHz.

Table 3-9 Typical baud rates and integer and fractional divisors

Programmed divisor (integer)	Programmed divisor (fraction)	Required bit rate (bps)	Generated bit rate (bps)	Error (%)
0x1	0x5	230400	231911	0.656
0x2	0xB	115200	115101	0.086
0x3	0x10	76800	76923	0.160
0x6	0x21	38400	38369	0.081
0x11	0x17	14400	14401	0.007
0x68	0xB	2400	2400	~0
0x8E0	0x2F	110	110	~0

3.3.7 Line control register, UARTLCR_H

The UARTLCR_H register is the line control register. This register accesses bits 29 to 22 of the PrimeCell UART bit rate and line control register, UARTLCR.

All the bits are cleared to 0 when reset. Table 3-10 shows the bit assignment of the UARTCR_H register.

Table 3-10 UARTLCR_H register

Bits	Name	Type	Function
15:8	-	-	Reserved, do not modify, read as zero.
7	Stick Parity Select (SPS)	Read/write	When bits 1, 2, and 7 of the UARTLCR_H register are set, the parity bit is transmitted and checked as a 0. When bits 1 and 7 are set, and bit 2 is 0, the parity bit is transmitted and checked as a 1. When this bit is cleared stick parity is disabled. Refer to Table 3-11 on page 3-14 for a truth table showing the SPS, EPS and PEN bits.
6:5	Word length [1:0] (WLEN)	Read/write	The select bits indicate the number of data bits transmitted or received in a frame as follows: 11 = 8 bits 10 = 7 bits 01 = 6 bits 00 = 5 bits.
4	Enable FIFOs (FEN)	Read/write	If this bit is set to 1, transmit and receive FIFO buffers are enabled (FIFO mode). When cleared to 0 the FIFOs are disabled (character mode) that is, the FIFOs become 1-byte-deep holding registers.
3	Two Stop Bits Select (STP2)	Read/write	If this bit is set to 1, two stop bits are transmitted at the end of the frame. The receive logic does not check for two stop bits being received.
2	Even Parity Select (EPS)	Read/write	If this bit is set to 1, even parity generation and checking is performed during transmission and reception, which checks for an even number of 1s in data and parity bits. When cleared to 0 then odd parity is performed which checks for an odd number of 1s. This bit has no effect when parity is disabled by Parity Enable (bit 1) being cleared to 0. Refer to Table 3-11 on page 3-14 for a truth table showing the SPS, EPS and PEN bits.
1	Parity Enable (PEN)	Read/write	If this bit is set to 1, parity checking and generation is enabled, else parity is disabled and no parity bit added to the data frame. Refer to Table 3-11 on page 3-14 for a truth table showing the SPS, EPS and PEN bits.
0	Send Break (BRK)	Read/write	If this bit is set to 1, a low-level is continually output on the UARTTXD output, after completing transmission of the current character. For the proper execution of the break command, the software must set this bit for at least two complete frames. For normal use, this bit must be cleared to 0.

UARTLCR_H, UARTIBRD and UARTFBRD form a single 30-bit wide register (UARTLCR) which is updated on a single write strobe generated by a UARTLCR_H write. So, in order to internally update the contents of UARTIBRD or UARTFBRD, a UARTLCR_H write must always be performed at the end.

———— **Note** ————

To update the three registers there are two possible sequences:

- UARTIBRD write, UARTFBRD write and UARTLCR_H write
- UARTFBRD write, UARTIBRD write and UARTLCR_H write.

To update UARTIBRD or UARTFBRD only:

- UARTIBRD write (or UARTFBRD write) and UARTLCR_H write.

Table 3-11 is a truth table for the SPS, EPS, and PEN bits of the UARTLCR_H register.

Table 3-11 Truth table

Parity Enable (PEN)	Even Parity Select (EPS)	Stick Parity Select (SPS)	Parity bit (transmitted or checked)
0	x	x	Not transmitted or checked
1	1	0	Even parity
1	0	0	Odd parity
1	0	1	1
1	1	1	0

———— **Note** ————

The baud rate and line control registers must not be changed:

- when the UART is enabled
- when completing a transmission or a reception when it has been programmed to become disabled.

The FIFO integrity is not guaranteed under the following conditions:

- after the BRK bit has been initiated
- if the software disables the UART in the middle of a transmission with data in the FIFO, and then re-enables it.

3.3.8 Control register, UARTCR

The UARTCR register is the control register. All the bits are cleared to 0 on reset except for bits 9 and 8 which are set to 1. Table 3-12 shows the bit assignment of the UARTCR register.

Table 3-12 UARTCR register

Bits	Name	Type	Function
15	CTS Hardware Flow Control Enable (CTSEn)	Read/write	If this bit is set to 1, CTS hardware flow control is enabled. Data is only transmitted when the nUARTCTS signal is asserted.
14	RTS Hardware Flow Control Enable (RTSEn)	Read/write	If this bit is set to 1, RTS hardware flow control is enabled. Data is only requested when there is space in the receive FIFO for it to be received.
13	Out2	Read/write	This bit is the complement of the PrimeCell UART Out2 (nUARTOut2) modem status output. That is, when the bit is programmed to a 1, the output is 0. For DTE this can be used as <i>Ring Indicator</i> (RI).
12	Out1	Read/write	This bit is the complement of the PrimeCell UART Out1 (nUARTOut1) modem status output. That is, when the bit is programmed to a 1 the output is 0. For DTE this can be used as <i>Data Carrier Detect</i> (DCD).
11	Request to Send (RTS)	Read/write	This bit is the complement of the PrimeCell UART request to send (nUARTRTS) modem status output. That is, when the bit is programmed to a 1, the output is 0.
10	Data Transmit Ready (DTR)	Read/write	This bit is the complement of the PrimeCell UART data transmit ready (nUARTDTR) modem status output. That is, when the bit is programmed to a 1, the output is 0.
9	Receive Enable (RXE)	Read/write	If this bit is set to 1, the receive section of the PrimeCell UART is enabled. Data reception occurs for either PrimeCell UART signals or SIR signals according to the setting of SIR Enable (bit 1). When the PrimeCell UART is disabled in the middle of reception, it completes the current character before stopping.
8	Transmit Enable (TXE)	Read/write	If this bit is set to 1, the transmit section of the PrimeCell UART is enabled. Data transmission occurs for either PrimeCell UART signals, or SIR signals according to the setting of SIR Enable (bit 1). When the PrimeCell UART is disabled in the middle of transmission, it completes the current character before stopping.

Table 3-12 UARTCR register (continued)

Bits	Name	Type	Function
7	Loop Back Enable (LBE)	Read/ write	If this bit is set to 1 and the SIR Enable bit is set to 1 and the test register UARTTCR bit 2 (SIRTEST) is set to 1, then the nSIROUT path is inverted, and fed through to the SIRIN path. The SIRTEST bit in the test register must be set to 1 to override the normal half-duplex SIR operation. This must be the requirement for accessing the test registers during normal operation, and SIRTEST must be cleared to 0 when loopback testing is finished. This feature reduces the amount of external coupling required during system test. If this bit is set to 1, and the SIRTEST bit is set to 0, the UARTTXD path is fed through to the UARTRXD path. In either SIR mode or normal mode, when this bit is set, the modem outputs are also fed through to the modem inputs. This bit is cleared to 0 on reset, which disables the loopback mode.
6:3	RESERVED	-	Reserved, do not modify, read as zero.
2	IrDA SIR Low Power Mode (SIRLP)	Read/ write	This bit selects the IrDA encoding mode. If this bit is cleared to 0, low-level bits are transmitted as an active high pulse with a width of $\frac{3}{16}$th of the bit period. If this bit is set to 1, low-level bits are transmitted with a pulse width which is 3 times the period of the IrLPBaud16 input signal, regardless of the selected bit rate. Setting this bit uses less power, but might reduce transmission distances.
1	SIR Enable (SIREN)	Read/ write	If this bit is set to 1, the IrDA SIR ENDEC is enabled. This bit has no effect if the UART is not enabled by bit 0 being set to 1. When the IrDA SIR ENDEC is enabled, data is transmitted and received on nSIROUT and SIRIN. UARTTXD remains in the marking state (set to 1). Signal transitions on UARTRXD or modem status inputs have no effect. When the IrDA SIR ENDEC is disabled, nSIROUT remains cleared to 0 (no light pulse generated), and signal transitions on SIRIN have no effect.
0	UART Enable (UARTEN)	Read/ write	If this bit is set to 1, the ARM PrimeCell UART is enabled. Data transmission and reception occurs for either PrimeCell UART signals or SIR signals according to the setting of SIR Enable (bit 1). When the PrimeCell UART is disabled in the middle of transmission or reception, it completes the current character before stopping.

———— **Note** ————

To enable transmission, both TXE, bit 8, and UARTEN, bit 0, must be set. Similarly, to enable reception, RXE, bit 9, and UARTEN, bit 0, must be set.

Note

Program the control registers as follows:

- 1. Disable the UART.
- 2. Wait for the end of transmission or reception of the current character.
- 3. Flush the transmit FIFO by disabling bit 4 (FEN) in the line control register (UARTCLR_H).
- 4. Reprogram the control register.
- 5. Enable the UART.

3.3.9 Interrupt FIFO level select register, UARTIFLS

The UARTIFLS register is the interrupt FIFO level select register. You can use the UARTIFLS register to define the FIFO level at which the **UARTTXINTR** and **UARTRXINTR** are triggered.

The interrupts are generated based on a transition through a level rather than being based on the level. That is, the design is such that the interrupts are generated when the fill level progresses through the trigger level.

The bits are reset so that the trigger level is when the FIFOs are at the half-way mark. Table 3-13 shows the bit assignment of the UARTIFLS register.

Table 3-13 UARTIFLS register

Bits	Name	Type	Function
15:6	-	-	Reserved, do not modify, read as zero.
5:3	Receive Interrupt FIFO Level Select (RXIFLSEL)	Read/write	The trigger points for the receive interrupt are as follows: 000 = Receive FIFO becomes >= 1/8 full 001 = Receive FIFO becomes >= 1/4 full 010 = Receive FIFO becomes >= 1/2 full 011 = Receive FIFO becomes >= 3/4 full 100 = Receive FIFO becomes >= 7/8 full 101:111 = reserved.
2:0	Transmit Interrupt FIFO Level Select (TXIFLSEL)	Read/write	The trigger points for the transmit interrupt are as follows: 000 = Transmit FIFO becomes <= 1/8 full 001 = Transmit FIFO becomes <= 1/4 full 010 = Transmit FIFO becomes <= 1/2 full 011 = Transmit FIFO becomes <= 3/4 full 100 = Transmit FIFO becomes <= 7/8 full 101:111 = reserved.

3.3.10 Interrupt mask set/clear register, UARTIMSC

The UARTIMSC register is the interrupt mask set/clear register. It is a read/write register.

On a read this register gives the current value of the mask on the relevant interrupt. On a write of 1 to the particular bit, it sets the corresponding mask of that interrupt. A write of 0 clears the corresponding mask.

All the bits are cleared to 0 when reset. Table 3-14 shows the bit assignment of the UARTIMSC register.

Table 3-14 UARTIMSC register

Bits	Name	Type	Function
15:11	RESERVED	-	Reserved, read as zero, do not modify.
10	Overrun Error Interrupt Mask (OEIM)	Read/write	On a read, the current mask for the OEIM interrupt is returned. On a write of 1, the mask of the OEIM interrupt is set. A write of 0 clears the mask.
9	Break Error Interrupt Mask (BEIM)	Read/write	On a read the current mask for the BEIM interrupt is returned. On a write of 1, the mask of the BEIM interrupt is set. A write of 0 clears the mask.
8	Parity Error Interrupt Mask (PEIM)	Read/write	On a read the current mask for the PEIM interrupt is returned. On a write of 1, the mask of the PEIM interrupt is set. A write of 0 clears the mask.
7	Framing Error Interrupt Mask (FEIM)	Read/write	On a read the current mask for the FEIM interrupt is returned. On a write of 1, the mask of the FEIM interrupt is set. A write of 0 clears the mask.
6	Receive Timeout Interrupt Mask (RTIM)	Read/write	On a read the current mask for the RTIM interrupt is returned. On a write of 1, the mask of the RTIM interrupt is set. A write of 0 clears the mask.
5	Transmit Interrupt Mask (TXIM)	Read/write	On a read the current mask for the TXIM interrupt is returned. On a write of 1, the mask of the TXIM interrupt is set. A write of 0 clears the mask.
4	Receive Interrupt Mask (RXIM)	Read/write	On a read the current mask for the RXIM interrupt is returned. On a write of 1, the mask of the RXIM interrupt is set. A write of 0 clears the mask.

Table 3-14 UARTIMSC register (continued)

Bits	Name	Type	Function
3	nUARTDSR Modem Interrupt Mask (DSRMIM)	Read/write	On a read the current mask for the DSRMIM interrupt is returned. On a write of 1, the mask of the DSRMIM interrupt is set. A write of 0 clears the mask.
2	nUARTDCD Modem Interrupt Mask (DCDMIM)	Read/write	On a read the current mask for the DCDMIM interrupt is returned. On a write of 1, the mask of the DCDMIM interrupt is set. A write of 0 clears the mask.
1	nUARTCTS Modem Interrupt Mask (CTSMIM)	Read/write	On a read the current mask for the CTSMIM interrupt is returned. On a write of 1, the mask of the CTSMIM interrupt is set. A write of 0 clears the mask.
0	nUARTRI Modem Interrupt Mask (RIMIM)	Read/write	On a read the current mask for the RIMIM interrupt is returned. On a write of 1, the mask of the RIMIM interrupt is set. A write of 0 clears the mask.

3.3.11 Raw interrupt status register, UARTRIS

The UARTRIS register is the raw interrupt status register. It is a read-only register. On a read this register gives the current raw status value of the corresponding interrupt. A write has no effect.

Caution

All the bits, except for the modem status interrupt bits (bits 3 to 0), are cleared to 0 when reset. The modem status interrupt bits are undefined after reset.

Table 3-15 shows the bit assignment of the UARTRIS register.

Table 3-15 UARTRIS register

Bits	Name	Type	Function
15:11	Reserved	-	Reserved, read as zero, do not modify
10	Overrun Error Interrupt Status (OERIS)	Read	Gives the raw interrupt state (prior to masking) of the UARTOEINTR interrupt
9	Break Error Interrupt Status (BERIS)	Read	Gives the raw interrupt state (prior to masking) of the UARTBEINTR interrupt

Table 3-15 UARTRIS register (continued)

Bits	Name	Type	Function
8	Parity Error Interrupt Status (PERIS)	Read	Gives the raw interrupt state (prior to masking) of the UARTPEINTR interrupt
7	Framing Error Interrupt Status (FERIS)	Read	Gives the raw interrupt state (prior to masking) of the UARTFEINTR interrupt
6	Receive Timeout Interrupt Status (RTRIS)	Read	Gives the raw interrupt state (prior to masking) of the UARTRTINTR interrupt ^a
5	Transmit Interrupt Status (TXRIS)	Read	Gives the raw interrupt state (prior to masking) of the UARTRXINTR interrupt
4	Receive Interrupt Status (RXRIS)	Read	Gives the raw interrupt state (prior to masking) of the UARTRXINTR interrupt
3	nUARTDSR Modem Interrupt Status (DSRRMIS)	Read	Gives the raw interrupt state (prior to masking) of the UARTDSRINTR interrupt
2	nUARTDCD Modem Interrupt Status (DCDRMIS)	Read	Gives the raw interrupt state (prior to masking) of the UARTDCDINTR interrupt
1	nUARTCTS Modem Interrupt Status (CTSRMIS)	Read	Gives the raw interrupt state (prior to masking) of the UARTCTSINTR interrupt
0	nUARTRI Modem Interrupt Status (RIRMIS)	Read	Gives the raw interrupt state (prior to masking) of the UARTRIINTR interrupt

- a. In this case the raw interrupt cannot be set unless the mask is set, this is because the mask acts as an enable for power saving. That is, the same status can be read from UARTMIS and UARTRIS for the receive timeout interrupt.

3.3.12 Masked interrupt status register, UARTMIS

The UARTMIS register is the masked interrupt status register. It is a read-only register. On a read this register gives the current masked status value of the corresponding interrupt. A write has no effect.

All the bits except for the modem status interrupt bits (bits 3 to 0) are cleared to 0 when reset. The modem status interrupt bits are undefined after reset. Table 3-16 shows the bit assignment of the UARTMIS register.

Table 3-16 UARTMIS register

Bits	Name	Type	Function
15:11	RESERVED	-	Reserved, read as zero, do not modify
10	Overrun Error Masked Interrupt Status (OEMIS)	Read	Gives the masked interrupt state (after masking) of the UARTOEINTR interrupt
9	Break Error Masked Interrupt Status (BEMIS)	Read	Gives the masked interrupt state (after masking) of the UARTBEINTR interrupt
8	Parity Error Masked Interrupt Status (PEMIS)	Read	Gives the masked interrupt state (after masking) of the UARTPEINTR interrupt
7	Framing Error Masked Interrupt Status (FEMIS)	Read	Gives the masked interrupt state (after masking) of the UARTFEINTR interrupt
6	Receive Timeout Masked Interrupt Status (RTMIS)	Read	Gives the masked interrupt state (after masking) of the UARTRTINTR interrupt
5	Transmit Masked Interrupt Status (TXMIS)	Read	Gives the masked interrupt state (after masking) of the UARTTXINTR interrupt
4	Receive Masked Interrupt Status (RXMIS)	Read	Gives the masked interrupt state (after masking) of the UARTRXINTR interrupt
3	nUARTDSR Modem Masked Interrupt Status (DSRMMIS)	Read	Gives the masked interrupt state (after masking) of the UARTDSRINTR interrupt
2	nUARTDCD Modem Masked Interrupt Status (DCDMMIS)	Read	Gives the masked interrupt state (after masking) of the UARTDCDINTR interrupt
1	nUARTCTS Modem Masked Interrupt Status (CTSMIS)	Read	Gives the masked interrupt state (after masking) of the UARTCTSINTR interrupt
0	nUARTRI Modem Masked Interrupt Status (RIMMIS)	Read	Gives the masked interrupt state (after masking) of the UARTRIINTR interrupt

3.3.13 Interrupt clear register, UARTICR

The UARTICR register is the interrupt clear register and is write-only. On a write of 1, the corresponding interrupt is cleared. A write of 0 has no effect. Table 3-17 shows the bit assignment of the UARTICR register.

Table 3-17 UARTICR register

Bits	Name	Type	Function
15:11	Reserved	-	Reserved, read as zero, do not modify
10	Overrun Error Interrupt Clear (OEIC)	Write	Clears the UARTOEINTR interrupt
9	Break Error Interrupt Clear (BEIC)	Write	Clears the UARTBEINTR interrupt
8	Parity Error Interrupt Clear (PEIC)	Write	Clears the UARTPEINTR interrupt
7	Framing Error Interrupt Clear (FEIC)	Write	Clears the UARTFEINTR interrupt
6	Receive Timeout Interrupt Clear (RTIC)	Write	Clears the UARTRTINTR interrupt
5	Transmit Interrupt Clear (TXIC)	Write	Clears the UARTTXINTR interrupt
4	Receive Interrupt Clear (RXIC)	Write	Clears the UARTRXINTR interrupt
3	nUARTDSR Modem Interrupt Clear (DSRMIC)	Write	Clears the UARTDSRINTR interrupt
2	nUARTDCD Modem Interrupt Clear (DCDMIC)	Write	Clears the UARTDCDINTR interrupt
1	nUARTCTS Modem Interrupt Clear (CTSMIC)	Write	Clears the UARTCTSINTR interrupt
0	nUARTRI Modem Interrupt Clear (RIMIC)	Write	Clears the UARTRIINTR interrupt

3.3.14 DMA control register, UARTDMACR

The UARTDMACR register is the DMA control register. It is a read/write register. All the bits are cleared to 0 on reset. Table 3-18 shows the bit assignment of the UARTDMACR register.

Table 3-18 UARTDMACR register

Bits	Name	Type	Function
15:3	RESERVED	-	Reserved, read as zero, do not modify.
2	DMA on Error (DMAONERR)	Read/ write	If this bit is set to 1, the DMA receive request outputs, UARTRXDMASREQ or UARTRXDMABREQ , are disabled when the UART error interrupt is asserted.
1	Transmit DMA Enable (TXDMAE)	Read/ write	If this bit is set to 1, DMA for the transmit FIFO is enabled.
0	Receive DMA Enable (RXDMAE)	Read/ write	If this bit is set to 1, DMA for the receive FIFO is enabled.

3.3.15 Peripheral identification registers, UARTPeriphID0-3

The UARTPeriphID0-3 registers are four 8-bit registers, that span address locations 0xFE0 - 0xFEC. The registers can conceptually be treated as a 32-bit register. The read only registers provide the following options of the peripheral:

PartNumber[11:0] This is used to identify the peripheral. The three digits product code 0x011 is used.

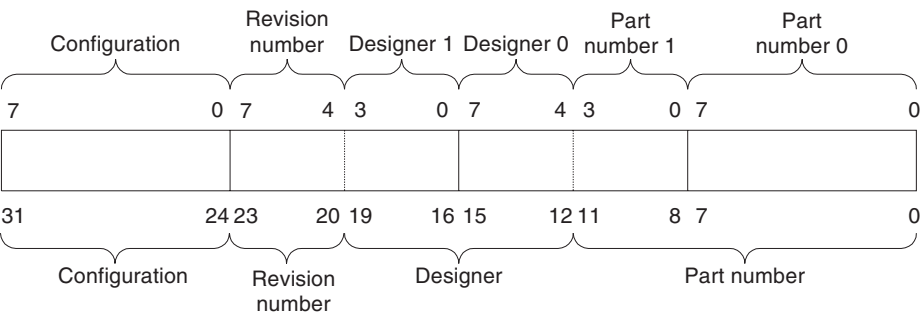
Designer ID[19:12] This is the identification of the designer. ARM Limited is 0x41 (ASCII A).

Revision[23:20] This is the revision number of the peripheral. The revision number starts from 0.

Configuration[31:24] This is the configuration option of the peripheral. The configuration value is 0.

Figure 3-1 on page 3-24 shows the bit assignment for the UARTPeriphID0-3 registers.

Actual register bit assignment



Conceptual register bit assignment

Figure 3-1 Peripheral identification register bit assignment

Note

When you design a systems memory map you must remember that the register has a 4KB memory footprint. All memory accesses to the peripheral identification registers must be 32-bit, using the LDR and STR instructions.

The four, 8-bit peripheral identification registers are described in the following subsections:

- *UARTPeriphID0 register*
- *UARTPeriphID1 register* on page 3-25
- *UARTPeriphID2 register* on page 3-25
- *UARTPeriphID3 register* on page 3-25.

UARTPeriphID0 register

The UARTPeriphID0 register is hard coded and the fields within the register determine the reset value. Table 3-19 shows the bit assignment of the UARTPeriphID0 register.

Table 3-19 UARTPeriphID0 register

Bits	Name	Description
15:8	-	Reserved, read undefined must read as zeros
7:0	PartNumber0	These bits read back as 0x11

UARTPeriphID1 register

The UARTPeriphID1 register is hard coded and the fields within the register determine the reset value. Table 3-20 shows the bit assignment of the UARTPeriphID1 register.

Table 3-20 UARTPeriphID1 register

Bits	Name	Description
15:8	-	Reserved, read undefined, must read as zeros
7:4	Designer0	These bits read back as 0x1
3:0	PartNumber1	These bits read back as 0x0

UARTPeriphID2 register

The UARTPeriphID2 register is hard coded and the fields within the register determine the reset value. Table 3-21 shows the bit assignment of the UARTPeriphID2 register.

Table 3-21 UARTPeriphID2 register

Bits	Name	Description
15:8	-	Reserved, read undefined, must read as zeros
7:4	Revision	These bits read back as 0x1
3:0	Designer1	These bits read back as 0x4

UARTPeriphID3 register

The UARTPeriphID3 register is hard coded and the fields within the register determine the reset value. Table 3-22 shows the bit assignment of the UARTPeriphID3 register.

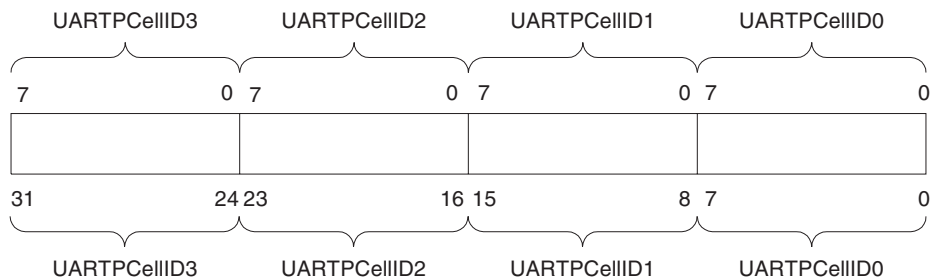
Table 3-22 UARTPeriphID3 register

Bits	Name	Description
15:8	-	Reserved, read undefined, must read as zeros
7:0	Configuration	These bits read back as 0x00

3.3.16 PrimeCell identification registers, UARTPCellID0-3

The UARTPCellID0-3 registers are four 8-bit wide registers, that span address locations 0xFF0-0xFFC. The registers can conceptually be treated as a 32-bit register. The register is used as a standard cross-peripheral identification system. The UARTPCellID register is set to 0xB105F00D. Figure 3-2 shows the bit assignment for the UARTPCellID0-3 registers.

Actual register bit assignment



Conceptual register bit assignment

Figure 3-2 PrimeCell identification register bit assignment

The four, 8-bit PrimeCell identification registers are described in the following subsections:

- *UARTPCellID0 register* on page 3-27
- *UARTPCellID1 register* on page 3-27
- *UARTPCellID2 register* on page 3-27
- *UARTPCellID3 register* on page 3-28.

UARTPCellID0 register

The UARTPCellID0 register is hard coded and the fields within the register determine the reset value. Table 3-23 shows the bit assignment of the UARTPCellID0 register.

Table 3-23 UARTPCellID0 register

Bits	Name	Description
15:8	-	Reserved, read undefined, must read as zeros
7:0	UARTPCellID0	These bits read back as 0x00

UARTPCellID1 register

The UARTPCellID1 register is hard coded and the fields within the register determine the reset value. Table 3-24 shows the bit assignment of the UARTPCellID1 register.

Table 3-24 UARTPCellID1 register read bits

Bits	Name	Description
15:8	-	Reserved, read undefined, must read as zeros
7:0	UARTPCellID1	These bits read back as 0xF0

UARTPCellID2 register

The UARTPCellID2 register is hard coded and the fields within the register determine the reset value. Table 3-25 shows the bit assignment of the UARTPCellID2 register.

Table 3-25 UARTPCellID2 register read bits

Bits	Name	Description
15:8	-	Reserved, read undefined, must read as zeros
7:0	UARTPCellID2	These bits read back as 0x05

UARTPCellID3 register

The UARTPCellID3 register is hard coded and the fields within the register determine the reset value. Table 3-26 shows the bit assignment of the UARTPCellID3 register.

Table 3-26 UARTPCellID3 register read bits

Bits	Name	Description
15:8	-	Reserved, read undefined, must read as zeros
7:0	UARTPCellID3	These bits read back as 0xB1

3.4 Interrupts

There are eleven maskable interrupts generated within the PrimeCell UART. These are combined to form five individual interrupt outputs and one which is the OR of the individual outputs:

- **UARTRXINTR**
- **UARTTXINTR.**
- **UARTRTINTR**
- **UARTMSINTR**, that can be caused by:
 - **UARTRIINTR**, due to change in the **nUARTRI** modem status line
 - **UARTCTSINTR**, due to change in the **nUARTCTS** modem status line
 - **UARTDCDINTR**, due to change in the **nUARTDCD** modem status line
 - **UARTDSRINTR**, due to change in the **nUARTDSR** modem status line.
- **UARTEINTR**, that can be caused by:
 - **UARTOEINTR**, due to an overrun error
 - **UARTBEINTR**, due to a break in the reception
 - **UARTPEINTR**, due to a parity error in the received character
 - **UARTFEINTR**, due to a framing error in the received character.
- **UARTINTR**, this is an OR function of the five individual masked outputs.

You can enable or disable the individual interrupts by changing the mask bits in the **UARTMSC** register. Setting the appropriate mask bit **HIGH** enables the interrupt.

Provision of individual outputs as well as a combined interrupt output, enables you to use either a global interrupt service routine, or modular device drivers to handle interrupts.

The transmit and receive dataflow interrupts **UARTRXINTR** and **UARTTXINTR** have been separated from the status interrupts. This enables you to use **UARTRXINTR** and **UARTTXINTR** so that data can be read or written in response to the FIFO trigger levels.

The error interrupt, **UARTEINTR**, can be triggered when there is an error in the reception of data. A number of error conditions are possible.

The modem status interrupt, **UARTMSINTR**, is a combined interrupt of all the individual modem status lines.

The status of the individual interrupt sources can be read either from **UARTRIS**, for raw interrupt status, or from the **UARTMIS**, for the masked interrupt status.

3.4.1 UARTMSINTR

The modem status interrupt is asserted if any of the modem status lines (**nUARTCTS**, **nUARTDCD**, **nUARTDSR**, and **nUARTRI**) change. It is cleared by writing a 1 to the corresponding bit(s) in the UARTICR register, depending on which of the modem status lines is causing the interrupt.

3.4.2 UARTRXINTR

The receive interrupt changes state when one of the following events occurs:

- If the FIFOs are enabled and the receive FIFO reaches the programmed trigger level. When this happens, the receive interrupt is asserted HIGH. The receive interrupt is cleared by reading data from the receive FIFO until it becomes less than the trigger level, or by clearing the interrupt.
- If the FIFOs are disabled (have a depth of one location) and data is received thereby filling the location, the receive interrupt is asserted HIGH. The receive interrupt is cleared by performing a single read of the receive FIFO, or by clearing the interrupt.

3.4.3 UARTTXINTR

The transmit interrupt changes state when one of the following events occurs:

- If the FIFOs are enabled and the transmit FIFO reaches the programmed trigger level. When this happens, the transmit interrupt is asserted HIGH. The transmit interrupt is cleared by writing data to the transmit FIFO until it becomes greater than the trigger level, or by clearing the interrupt.
- If the FIFOs are disabled (have a depth of one location) and there is no data present in the transmitters single location, the transmit interrupt is asserted HIGH. It is cleared by performing a single write to the transmit FIFO, or by clearing the interrupt.

To update the transmit FIFO you must:

- Write data to the transmit FIFO, either prior to enabling the PrimeCell UART and the interrupts, or after enabling the PrimeCell UART and interrupts.

———— Note ————

The transmit interrupt is based on a transition through a level, rather than on the level itself. When the interrupt and the PrimeCell UART is enabled before any data is written to the transmit FIFO the interrupt is not set. The interrupt is only set once written data leaves the single location of the transmit FIFO and it becomes empty.

3.4.4 UARTRTINTR

The receive timeout interrupt is asserted when the receive FIFO is not empty, and no further data is received over a 32-bit period. The receive timeout interrupt is cleared either when the FIFO becomes empty through reading all the data (or by reading the holding register), or when a 1 is written to the corresponding bit of the UARTICR register.

3.4.5 UARTEINTR

The error interrupt is asserted when an error occurs in the reception of data by the PrimeCell UART. The interrupt can be caused by a number of different error conditions:

- framing
- parity
- break
- overrun.

You can determine the cause of the interrupt by reading the UARTRIS or UARTMIS registers. It can be cleared by writing to the relevant bits of the UARTICR register (bits 7 to 10 are the error clear bits).

3.4.6 UARTINTR

The interrupts are also combined into a single output, which is an OR function of the individual masked sources. You can connect this output to the system interrupt controller to provide another level of masking on a individual peripheral basis.

The combined PrimeCell UART interrupt is asserted if any of the individual interrupts above are asserted and enabled.

Chapter 4

Programmer's Model for Test

This chapter describes the additional logic for integration testing. It contains the following sections:

- *PrimeCell UART test harness overview* on page 4-2
- *Scan testing* on page 4-3
- *Test registers* on page 4-4
- *Integration testing of block inputs* on page 4-10
- *Integration testing of block outputs* on page 4-12
- *Integration test summary* on page 4-15.

4.1 PrimeCell UART test harness overview

The additional logic for functional verification and integration vectors enables:

- capture of input signals to the block
- stimulation of the output signals.

The integration vectors provide a way of verifying that the PrimeCell UART is correctly wired into a system. This is done by separately testing three groups of signals:

AMBA signals

These are tested by checking the connections of all the address and data bits.

Primary input/output signals

These are tested using a simple trickbox that can demonstrate the correct connection of the input/output signals to external pads.

Intra-chip signals (such as interrupt sources)

The tests for these signals are system-specific, and enable you to write the necessary tests. Additional logic is implemented enabling you to read and write to each intra-chip input/output signal.

These test features are controlled by test registers. This enables you to test the PrimeCell UART in isolation from the rest of the system using only transfers from the AMBA APB.

Off-chip test vectors are supplied using a 32-bit parallel *External Bus Interface* (EBI) and converted to internal AMBA bus transfers. The application of test vectors is controlled through the *Test Interface Controller* (TIC) AMBA bus master module.

4.2 Scan testing

The PrimeCell UART has been designed to simplify:

- insertion of scan test cells
- use of *Automatic Test Pattern Generation* (ATPG).

This provides an alternative method of manufacturing test.

4.3 Test registers

The PrimeCell UART test registers are memory-mapped as shown in Table 4-1.

Table 4-1 Test registers memory map

Address	Type	Width	Reset value	Name	Description
UartBase + 0x080	Read/write	3	0x0	UARTTCR	Test control register
UartBase + 0x084	Read/write	8	0x00	UARTITIP	Integration test input read/set register
UartBase + 0x088	Read/write	14	0x000	UARTITOP	Integration test output read/set register
UartBase + 0x08C	Read/write	11	0x---	UARTTDR	Test data register

Each register shown in Table 4-1 is described in the following sections:

- *Test control register, UARTTCR* on page 4-5
- *Integration test input read/set register, UARTITIP* on page 4-6
- *Integration test output read/set register, UARTITOP* on page 4-7
- *Test data register, UARTTDR* on page 4-9.

4.3.1 Test control register, UARTTCR

UARTTCR is the test control register. This general test register controls operation of the PrimeCell UART under test conditions. Table 4-2 shows the bit assignments for the UARTTCR.

Table 4-2 UARTTCR register bits

Bits	Name	Description
16:3	-	Reserved, unpredictable when read.
2	SIR Test Enable (SIRTEST)	Setting this bit to 1 enables the receive data path during IrDa transmission (testing requires the SIR to be configured in full-duplex mode). This bit must be set to 1 to enable SIR system loop back testing, when the normal mode control register UARTCR bit 7, <i>Loop Back Enable</i> (LBE) has been set to 1. Clearing this bit to 0 disables the receive logic when the SIR is transmitting (normal operation). This bit defaults to 0 for normal operation (half-duplex operation).
1	Test fifo enable (TESTFIFO)	When this bit is 1, a write to the UARTTDR writes data into the receive FIFO, and reads from the UARTTDR reads data out of the transmit FIFO. When this bit is 0, data cannot be read directly from the transmit FIFO or written directly to the receive FIFO (normal operation). The reset value is 0.
0	ITEN	Integration test enable. When this bit is 1, the UART is placed in integration test mode, otherwise it is in normal mode.

4.3.2 Integration test input read/set register, UARTITIP

UARTITIP is the integration test input read/set register. It is a a read/write register. In integration test mode it enables inputs to be both written to and read from. Table 4-3 shows the bit assignments for the UARTITIP.

Table 4-3 UARTITIP register bits

Bits	Name	Description
16:8	-	Reserved, unpredictable when read.
7	UARTTXDMACLR	Writes to this bit specify the value to be driven on the intra-chip input, UARTTXDMACLR , in the integration test mode. Reads return the value of UARTTXDMACLR at the output of the test multiplexor.
6	UARTRXDMACLR	Writes to this bit specify the value to be driven on the intra-chip input, UARTRXDMACLR , in the integration test mode. Reads return the value of UARTRXDMACLR at the output of the test multiplexor.
5	nUARTRI	Reads return the value of the nUARTRI primary input.
4	nUARTDCD	Reads return the value of the nUARTDCD primary input.
3	nUARTCTS	Reads return the value of the nUARTCTS primary input.
2	nUARTDSR	Reads return the value of the nUARTDSR primary input.
1	SIRIN	Reads return the value of the SIRIN primary input.
0	UARTRXD	Reads return the value of the UARTRXD primary input.

4.3.3 Integration test output read/set register, UARTITOP

UARTITOP is the Integration test output read/set register. The primary outputs are read only and the intra-chip outputs are read/write. In integration test mode it enables outputs to be both written to and read from. Table 4-4 shows the bit assignments for the UARTITOP.

Table 4-4 UARTITOP register bits

Bits	Name	Description
15	UARTTXDMASREQ	Intra-chip output. Writes specify the value to be driven on the UARTTXDMASREQ line in the integration test mode. Reads return the value of UARTTXDMASREQ at the output of the test multiplexor.
14	UARTTXDMABREQ	Intra-chip output. Writes specify the value to be driven on the UARTTXDMABREQ line in the integration test mode. Reads return the value of UARTTXDMABREQ at the output of the test multiplexor.
13	UARTRXDMASREQ	Intra-chip output. Writes specify the value to be driven on the UARTRXDMASREQ line in the integration test mode. Reads return the value of UARTRXDMASREQ at the output of the test multiplexor.
12	UARTRXDMABREQ	Intra-chip output. Writes specify the value to be driven on the UARTDMABREQ line in the integration test mode. Reads return the value of UARTDMABREQ at the output of the test multiplexor.
11	UARTMSINTR	Intra-chip output. Writes specify the value to be driven on the UARTMSINTR line in the integration test mode. Reads return the value of UARTMSINTR at the output of the test multiplexor.
10	UARTRXINTR	Intra-chip output. Writes specify the value to be driven on the UARTRXINTR line in the integration test mode. Reads return the value of UARTRXINTR at the output of the test multiplexor.

Table 4-4 UARTITOP register bits (continued)

Bits	Name	Description
9	UARTTXINTR	Intra-chip output. Writes specify the value to be driven on the UARTTXINTR line in the integration test mode. Reads return the value of UARTTXINTR at the output of the test multiplexor.
8	UARTRTINTR	Intra-chip output. Writes specify the value to be driven on the UARTRTINTR line in the integration test mode. Reads return the value of UARTRTINTR at the output of the test multiplexor.
7	UARTEINTR	Intra-chip output. Writes specify the value to be driven on the UARTEINTR line in the integration test mode. Reads return the value of UARTEINTR at the output of the test multiplexor.
6	UARTINTR	Intra-chip output. Writes specify the value to be driven on the UARTINTR line in the integration test mode. Reads return the value of UARTINTR at the output of the test multiplexor.
5	nUARTOut2	Primary output. Writes specify the value to be driven on the nUARTOut2 line in the integration test mode.
4	nUARTOut1	Primary output. Writes specify the value to be driven on the nUARTOut1 line in the integration test mode.
3	nUARTRTS	Primary output. Writes specify the value to be driven on the nUARTRTS line in the integration test mode.
2	nUARTDTR	Primary output. Writes specify the value to be driven on the nUARTDTR line in the integration test mode.
1	nSIROUT	Primary output. Writes specify the value to be driven on the nSIROUT line in the integration test mode.
0	UARTTXD	Primary output. Writes specify the value to be driven on the UARTTXD line in the integration test mode.

4.3.4 Test data register, UARTTDR

UARTTDR is the test data register. It enables data to be written into the receive FIFO and read out from the transmit FIFO for test purposes. This test function is enabled by the **TESTFIFO** signal, bit 1 of the test control register (UARTTCR). Table 4-5 shows the bit assignments for the UARTTDR.

Table 4-5 UARTTDR register bits

Bits	Name	Description
16:11	-	Reserved, unpredictable when read
10:0	DATA	When the TESTFIFO signal is asserted, data is written into the receive FIFO and read out of the transmit FIFO

4.4 Integration testing of block inputs

The following sections describe the integration testing for the block inputs:

- *Intra-chip inputs*
- *Primary inputs* on page 4-11.

4.4.1 Intra-chip inputs

Figure 4-1 explains the implementation details of the input integration test harness. The ITEN bit is used as the control bit for the multiplexor, which is used in the read path of the **UARTTXDMACLR** and **UARTRXDMACLR** intra-chip inputs. If the ITEN control bit is deasserted, the **UARTTXDMACLR** and **UARTRXDMACLR** intra-chip inputs are routed as the internal **UARTTXDMACLR** and **UARTRXDMACLR** inputs respectively, otherwise the stored register values are driven on the internal line. All other read only bits in the UARTITIP register are connected directly to the primary input pins.

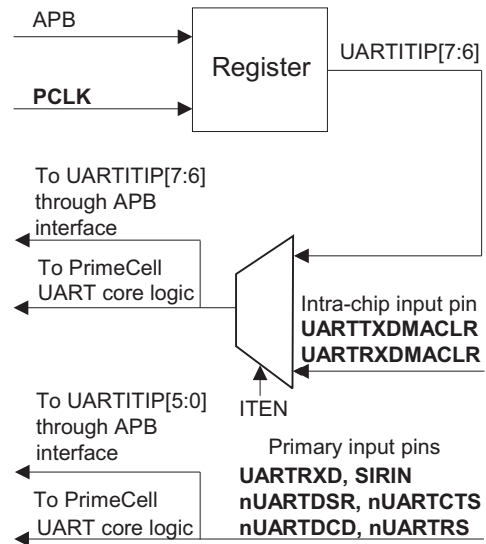


Figure 4-1 Input integration test harness

When you run integration tests with the PrimeCell UART in a standalone test setup:

- Write a 1 to the ITEN bit in the control register. This selects the test path from the UARTITIP[7:6] register bits to the **UARTRXDMACLR** and **UARTTXDMACLR** signals.

- Write a 1 and then a 0 to each of the UARTITIP[7:6] register bits, and read the same register bits to ensure that the value written is read out.

When you run integration tests with the PrimeCell UART as part of an integrated system:

- Write a 0 to the ITEN bit in the control register. This selects the normal path from the external **UARTRXDMACLR** pin to the internal **UARTRXDMACLR** signal, and the path from the external **UARTTXDMACLR** pin to the internal **UARTTXDMACLR** pin.
- Write a 1 and then a 0 to the internal test registers of the DMA controller to toggle the **UARTRXDMACLR** signal connection between the DMA controller and the PrimeCell UART. Read from the UARTITIP[6] register bit to verify that the value written into the DMA controller, is read out through the PrimeCell UART. Similarly, write a 1 and then a 0 to the internal registers of the DMA controller to toggle the **UARTTXDMACLR** signal connection between the DMA controller and the PrimeCell UART. Read from the UARTITIP[7] register bit to verify that the value written into the DMA controller, is read out through the PrimeCell UART.

4.4.2 Primary inputs

The primary inputs are tested using the integration vector trickbox by looping back primary outputs as follows:

- **UARTTXD** to **UARTRXD**
- **nSIROUT** to **SIRIN**
- **nUARTRTS** to **nUARTCTS**
- **nUARTOUT1** to **nUARTDCD**
- **nUARTDTR** to **nUARTDSR**
- **nUARTOUT2** to **nUARTRI**.

Write a 1 to the ITEN bit in the control register. 1s and 0s are driven onto the primary output lines through the UARTITOP register bits [5:0], and read back through the UARTITIP register bits [5:0].

4.5 Integration testing of block outputs

The following sections describe the integration testing for the block outputs:

- *Intra-chip outputs*
- *Primary outputs* on page 4-13.

4.5.1 Intra-chip outputs

Use this test for the following outputs:

- **UARTTXDMASREQ**
- **UARTTXDMABREQ**
- **UARTRXDMASREQ**
- **UARTRXDMABREQ**
- **UARTINTR**
- **UARTMSINTR**
- **UARTRXINTR**
- **UARTTXINTR**
- **UARTRTINTR**
- **UARTEINTR**.

When you run integration tests with the PrimeCell UART in a standalone test setup:

- Write a 1 to the ITEN bit in the control register. This selects the test path from the UARTITOP0[15:6] register bits to the intra-chip output signals.
- Write a 1 and then a 0 to the UARTITOP0[15:6] register bits, and read the same register bits to verify that the value written is read out.

When you run integration tests with the PrimeCell UART as part of an integrated system:

- Write a 1 to the ITEN bit in the control register. This selects the test path from the UARTITOP0[15:6] register bits to the intra-chip output signals.
- Write a 1 and then a 0 to the UARTITOP0[15:6] register bits to toggle the signal connections between the DMA controller/interrupt controller and the PrimeCell UART. Read from the internal test registers of the DMA controller/interrupt controller to verify that the value written into the UARTITOP0[15:6] register bits is read out through the PrimeCell UART.

Figure 4-2 on page 4-13 explains the implementation details of the output integration test harness for intra-chip outputs.

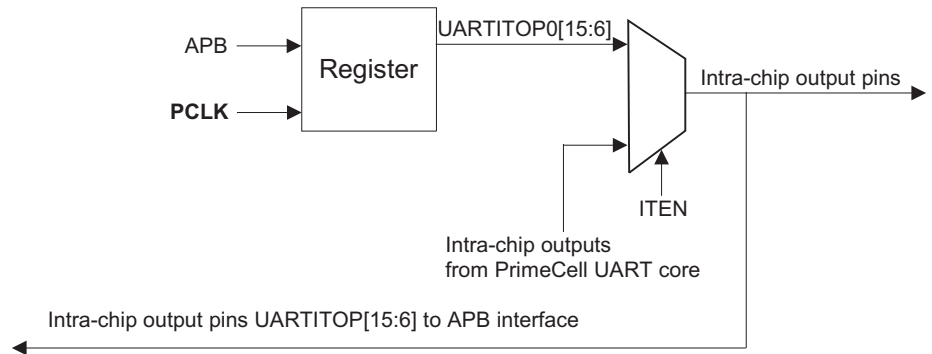


Figure 4-2 Output integration test harness, intra-chip outputs

4.5.2 Primary outputs

Integration testing of primary outputs and primary inputs is carried out using the integration vector trickbox. Use this test for the following outputs:

- **UARTTXD**
- **nSIROUT**
- **nUARTDTR**
- **nUARTRTS**
- **nUARTOut1**
- **nUARTOut2.**

Verify the primary input and output pin connections as follows:

The primary output pins (listed above) are looped back to the primary input pins through the integration vector trickbox.

- All the primary outputs can be accessed through the UARTITOP register. Different data patterns are written to the output pins using the UARTITOP registers.
- The looped back data is read back through the UARTTIP register.

Figure 4-3 on page 4-14 explains the implementation details of the output integration test harness in the case of primary outputs.

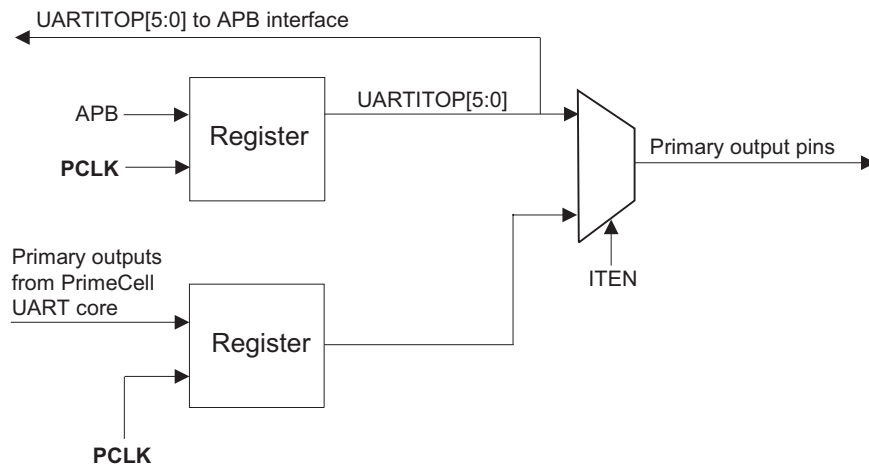


Figure 4-3 Output integration test harness, primary outputs

4.6 Integration test summary

Table 4-6 summarizes the integration test strategy for all PrimeCell UART pins.

Table 4-6 PrimeCell UART integration test strategy

Name	Type	Source/ destination	Test strategy
PRESETn	Input	Reset controller	Not tested using integration test vectors
PADDR [11:2]	Input	APB	Register read/write
PCLK	Input	APB	Register read/write
PENABLE	Input	APB	Register read/write.
PRDATA [15:0]	Output	APB	Register read/write
PSEL	Input	APB	Register read/write
PWDATA [15:0]	Input	APB	Register read/write
PWRITE	Input	APB	Register read/write
UARTCLK	Input	Clock generator	Not tested using integration test vectors
nUARTRST	Input	Reset controller	Not tested using integration test vectors
UARTMSINTR	Output	Interrupt controller	Using UARTITOP register
UARTRXINTR	Output	Interrupt controller	Using UARTITOP register
UARTTXINTR	Output	Interrupt controller	Using UARTITOP register
UARTRTINTR	Output	Interrupt controller	Using UARTITOP register
UARTEINTR	Output	Interrupt controller	Using UARTITOP register
UARTTXDMASREQ	Output	DMA controller	Using UARTITOP register
UARTRXDMASREQ	Output	DMA controller	Using UARTITOP register
UARTTXDMABREQ	Output	DMA controller	Using UARTITOP register
UARTRXDMABREQ	Output	DMA controller	Using UARTITOP register
UARTTXDMACLR	Input	DMA controller	Using UARTITIP register
UARTRXDMACLR	Input	DMA controller	Using UARTITIP register
UARTINTR	Output	Interrupt controller	Using UARTITOP register

Table 4-6 PrimeCell UART integration test strategy (continued)

Name	Type	Source/ destination	Test strategy
SCANENABLE	Input	Test controller	Not tested using integration test vectors
SCANINPCLK	Input	Test controller	Not tested using integration test vectors
SCANINUCLK	Input	Test controller	Not tested using integration test vectors
SCANOUTPCLK	Output	Test controller	Not tested using integration test vectors
SCANOUTUCLK	Output	Test controller	Not tested using integration test vectors
nUARTCTS	Input	PAD	Using integration vector trickbox and UARTITIP and UARTITOP registers
nUARTDCD	Input	PAD	Using integration vector trickbox and UARTITIP and UARTITOP registers
nUARTDSR	Input	PAD	Using integration vector trickbox and UARTITIP and UARTITOP registers
nUARTRI	Input	PAD	Using integration vector trickbox and UARTITIP and UARTITOP registers
UARTRXD	Input	PAD	Using integration vector trickbox and UARTITIP and UARTITOP registers
SIRIN	Input	PAD	Using integration vector trickbox and UARTITIP and UARTITOP registers
UARTTXD	Output	PAD	Using integration vector trickbox and UARTITIP and UARTITOP registers
nSIROUT	Output	PAD	Using integration vector trickbox and UARTITIP and UARTITOP registers
nUARTDTR	Output	PAD	Using integration vector trickbox and UARTITIP and UARTITOP registers
nUARTRTS	Output	PAD	Using integration vector trickbox and UARTITIP and UARTITOP registers
nUARTOut1	Output	PAD	Using integration vector trickbox and UARTITIP and UARTITOP registers
nUARTOut2	Output	PAD	Using integration vector trickbox and UARTITIP and UARTITOP registers

Appendix A

ARM PrimeCell UART (PL011) Signal Descriptions

This appendix describes the signals that interface with the ARM PrimeCell UART (PL011) block. It contains the following sections:

- *AMBA APB signals* on page A-2
- *On-chip signals* on page A-3
- *Signals to pads* on page A-5.

A.1 AMBA APB signals

The PrimeCell UART module is connected to the AMBA APB as a bus slave. AMBA APB signals have a **P** prefix and are active HIGH. Active LOW signals contain a lower case **n**. The AMBA APB signals are described in Table A-1.

Table A-1 AMBA APB signal descriptions

Name	Type	Source/ destination	Description
PRESETn	Input	Reset controller	Bus reset signal, active LOW.
PADDR [11:2]	Input	APB	Subset of AMBA APB address bus.
PCLK	Input	APB	AMBA APB clock, used to time all bus transfers.
PENABLE	Input	APB	AMBA APB enable signal. PENABLE is asserted HIGH for one cycle of PCLK to enable a bus transfer.
PRDATA [15:0]	Output	APB	Subset of unidirectional AMBA APB read data bus.
PSEL	Input	APB	PrimeCell UART and SIR ENDEC select signal from decoder. When set to 1 this signal indicates the slave device is selected by the AMBA APB bridge, and that a data transfer is required.
PWDATA [15:0]	Input	APB	Subset of unidirectional AMBA APB write data bus.
PWRITE	Input	APB	AMBA APB transfer direction signal, indicates a write access when HIGH, read access when LOW.

A.2 On-chip signals

A free-running reference clock, **UARTCLK**, must be provided. By default it is assumed to be asynchronous to **PCLK**. The **UARTCLK** clock must have a frequency between 1.42MHz to 542.72MHz to ensure that the low-power mode transmit pulse duration complies with the IrDA SIR specification.

The reset inputs are asynchronously asserted but synchronously removed for each of the clock domains within the PrimeCell UART. This ensures that logic is reset even if clocks are not present, to avoid any static power consumption problems at power up. Each clock domain has a individual reset to simplify the process of inserting scan test cells.

The on-chip signals required in addition to the AMBA APB signals are shown in Table A-2.

Table A-2 On-chip signal descriptions

Name	Type	Source/ destination	Description
UARTCLK	Input	Clock generator	PrimeCell UART reference clock.
nUARTRST	Input	Reset controller	PrimeCell UART reset signal to UARTCLK clock domain, active LOW. The reset controller must use PRESETn to assert nUARTRST asynchronously but negate it synchronously with UARTCLK .
UARTMSINTR	Output	Interrupt controller	PrimeCell UART modem status interrupt (active HIGH).
UARTRXINTR	Output	Interrupt controller	PrimeCell UART receive FIFO interrupt (active HIGH).
UARTTXINTR	Output	Interrupt controller	PrimeCell UART transmit FIFO interrupt (active HIGH).
UARTRTINTR	Output	Interrupt controller	PrimeCell UART receive timeout interrupt (active HIGH).
UARTEINTR	Output	Interrupt controller	PrimeCell UART error interrupt (active HIGH).
UARTINTR	Output	Interrupt controller	PrimeCell UART interrupt (active HIGH). A single combined interrupt generated as an OR function of the five individually maskable interrupts above.
UARTTXDMASREQ	Output	DMA controller	PrimeCell UART transmit DMA single request (active HIGH).
UARTRXDMASREQ	Output	DMA controller	PrimeCell UART receive DMA single request (active HIGH).

Table A-2 On-chip signal descriptions (continued)

Name	Type	Source/ destination	Description
UARTTXDMABREQ	Output	DMA controller	PrimeCell UART transmit DMA burst request (active HIGH).
UARTRXDMABREQ	Output	DMA controller	PrimeCell UART receive DMA burst request (active HIGH).
UARTTXDMACLR	Input	DMA controller	DMA request clear, asserted by the DMA controller to clear the transmit request signals. If DMA burst transfer is requested, the clear signal is asserted during the transfer of the last data in the burst.
UARTRXDMACLR	Input	DMA controller	DMA request clear, asserted by the DMA controller to clear the receive request signals. If DMA burst transfer is requested, the clear signal is asserted during the transfer of the last data in the burst.
SCANENABLE	Input	Test controller	PrimeCell UART scan enable signal for both clock domains.
SCANINPCLK	Input	Test controller	PrimeCell UART input scan signal for the PCLK domain.
SCANINUCLK	Input	Test controller	PrimeCell UART input scan signal for the UARTCLK domain.
SCANOUTPCLK	Output	Test controller	PrimeCell UART output scan signal for the PCLK domain.
SCANOUTUCLK	Output	Test controller	PrimeCell UART output scan signal for the UARTCLK domain.

A.3 Signals to pads

Table A-3 describes the signals from the PrimeCell UART and IrDA SIR ENDEC to input/output pads of the chip. You must make proper use of the peripheral pins to meet the exact interface requirements.

Table A-3 Pad signal descriptions

Name	Type	Pad type	Description
nUARTCTS	Input	PAD	PrimeCell UART Clear To Send modem status input, active LOW. The condition of this signal can be read from the UARTFR register.
nUARTDCD	Input	PAD	PrimeCell UART Data Carrier Detect modem status input, active LOW. The condition of this signal can be read from the UARTFR register.
nUARTDSR	Input	PAD	PrimeCell UART Data Set Ready modem status input, active LOW. The condition of this signal can be read from the UARTFR register.
nUARTRI	Input	PAD	PrimeCell UART Ring Indicator modem status input, active LOW. The condition of this signal can be read from the UARTFR register.
UARTRXD	Input	PAD	PrimeCell UART Received Serial Data input.
SIRIN	Input	PAD	SIR Received Serial Data Input. In the idle state, the signal remains in the marking state 1. When a light pulse is received which represents a logic 0, this signal is a 0.
UARTTXD	Output	PAD	PrimeCell UART Transmitted Serial Data output. Defaults to the marking state 1, when reset.
nSIROUT	Output	PAD	SIR Transmitted Serial Data Output, active LOW. In the idle state, this signal remains 0 (the marking state). When this signal is set to 1, an infrared light pulse is generated which represents a logic 0 (spacing state).
nUARTDTR	Output	PAD	PrimeCell UART Data Terminal Ready modem status output, active LOW. The reset value is 0.

Table A-3 Pad signal descriptions (continued)

Name	Type	Pad type	Description
nUARTRTS	Output	PAD	PrimeCell UART Request to Send modem status output, active LOW. The reset value is 0.
nUARTOut1	Output	PAD	PrimeCell UART Out1 modem status output, active LOW. The reset value is 0.
nUARTOut2	Output	PAD	PrimeCell UART Out2 modem status output, active LOW. The reset value is 0.

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